Review of Last Lecture

6.1 Introduction

6.2 Discrete-Time Signals – Sequences

6.3 Mathematical Model of Discrete-Time Systems

6.4 Classical Time-Domain Solution of Linear Constant-Coefficient Difference Equations

6.5 Zero-State Response and Zero-Input Response of Discrete-Time Systems

6.6 Unit Sample Response of Discrete-Time Systems

6.7 Convolution Sum

Contents of This Lecture

Learning Objectives of This Lecture

1. Be familiar with the judgment and period calculation of discrete periodic signals;
2. Be familiar with the basic operations of sequences;
3. Be able to establish the difference equation of discrete-time systems;
4. Be familiar with the classical time-domain solution method of difference equations and correctly use **boundary conditions**.

6.1 Introduction

Continuous-Time Signals:

$f(t)$ is a continuously changing function of t, and a definite function value can be given for any time value t except for several discontinuous points. The waveform of the signal is a smooth curve, and it is generally also called an analog signal.

Continuous-Time Systems:

The input and output of the system are both continuous-time signals, e.g., physics, modern circuit theory, and analog communication systems.

单选题 1分

设置

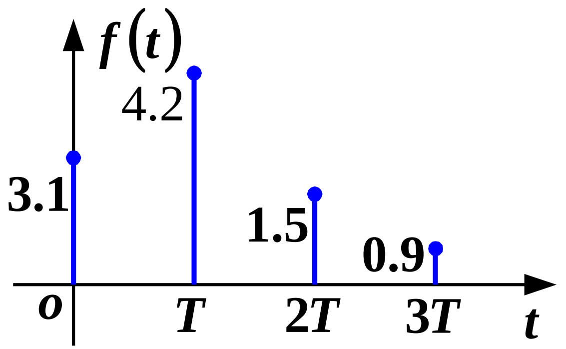
True or False: The signal obtained by sampling a continuous-time signal is a digital signal.

A True

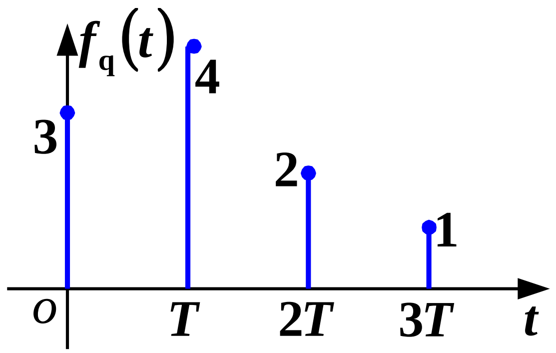
B False

提交

Discrete-Time Signal: The time variable is discrete, and the function has definite values only at certain specified moments, being undefined at other times. It can be obtained by sampling an analog signals or generated by an actual system.



Sampling Process — Obtains discrete-time signals.



Amplitude Quantization — The amplitude is of limited levels.

Digital Signal: A signal where the amplitude of a discrete-time signal is quantized at each discrete point.

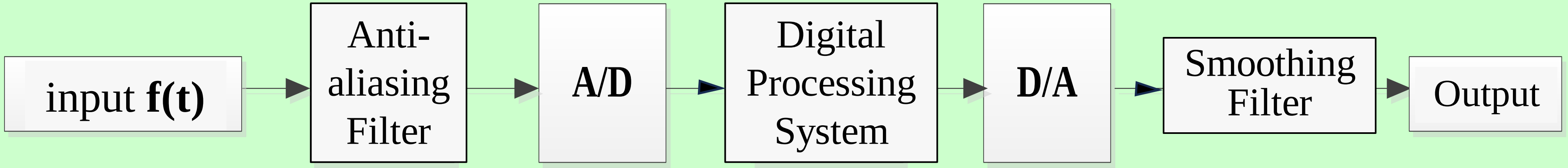
6.1 Introduction

Discrete-Time System: Both the input and output of the system are discrete-time signals. Examples include computer, numerical analysis, statistics, and economics.

Advantages of Discrete-Time Systems：

1. Facilitate large-scale integration, featuring small size, light weight, and low power consumption;
2. High reliability and high precision (depending on the number of bits of binary numbers);
3. Store information using memory, with flexible functions;
4. Easy to eliminate noise and process low-frequency signals;
5. Process multi-dimensional signals;
6. Programmable technology improves flexibility and versatility.

Hybrid System: The combined application of continuous-time systems and discrete-time systems. Examples include automatic control systems and digital communication systems.



Signal Processing System

6.1 Introduction

Continuous-Time System	vs.	Discrete-Time System
• Differential Equation		• Difference Equation
• Convolution Integral		• Convolution Sum
• Laplace Transform		• z-Transform
• Fourier Transform		• Discrete-Time Fourier Transform, Discrete Fourier Transform
• System Function		• System Function
• Convolution Theorem		• Convolution Theorem

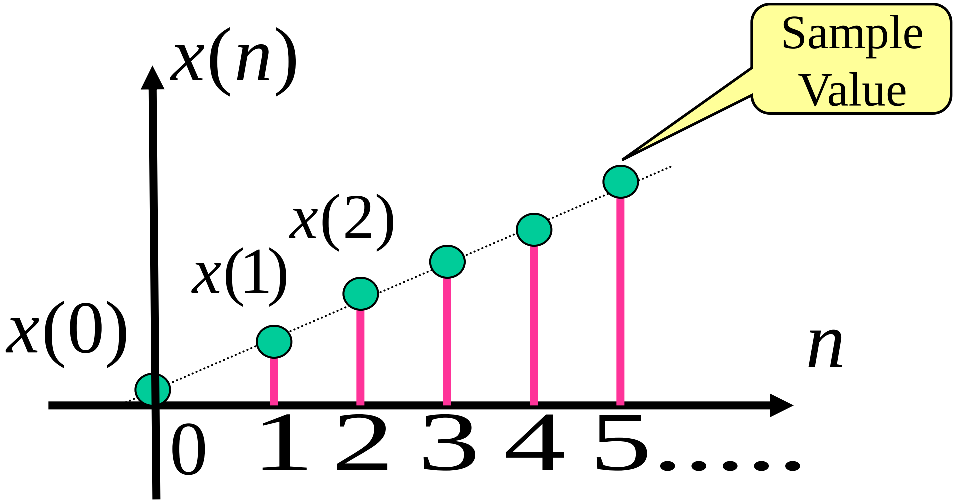
Discrete-Time Signal (Sequence): It is discrete in time, with function values defined only at certain discontinuous specified moments and undefined at other times.

The interval between discrete moments is uniform, denoted as *T*.

Discrete-time signals are denoted as $\{x(nT)\}$ or $\{x(n)\}$, $n \in Z, -\infty < n < \infty$

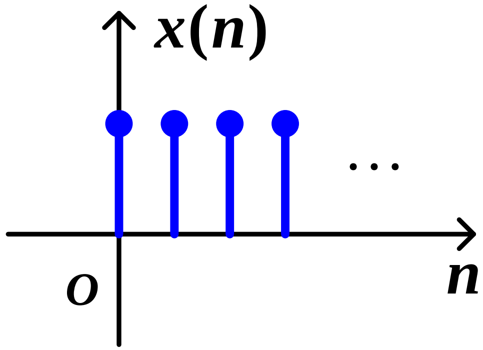
For simplicity, the sequence is denoted as ***x*(*n*).**

- Listing values, e.g., ***x*(*n*)={1,2,3}**
- Closed-form solution, e.g., $\boldsymbol{x(n)=a^n u(n)}$
- Graphical representation, as shown in the figure

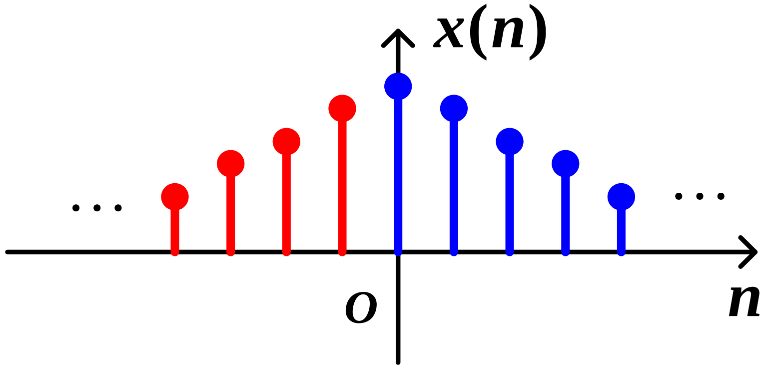


Three Forms of Sequences：

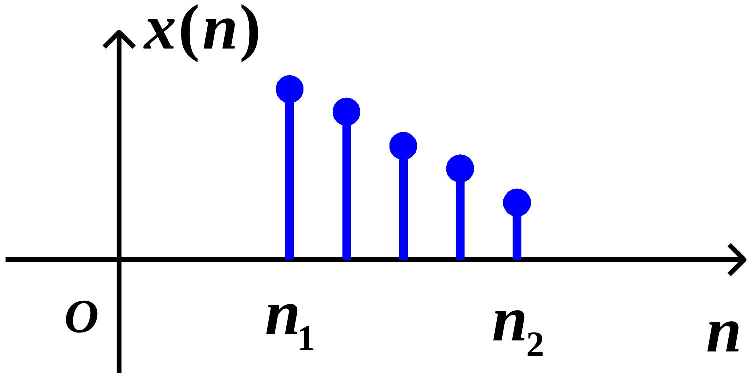
Single-sided (Causal) Sequence： $n \geq 0$;



Double-sided Sequence：$-\infty \leq n \leq \infty$;



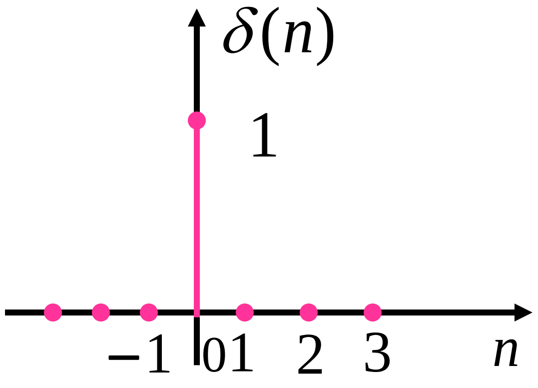
Finite-Length Sequence： $n_1 \leq n \leq n_2$.



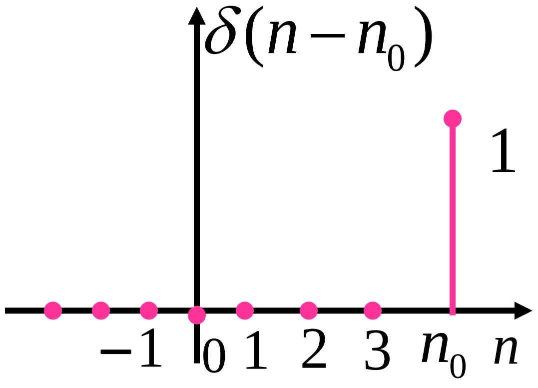
6.2.1 Commonly Used Typical Sequences

1. Unit Sample/ Unit Impulse

$$\delta(n)=\begin{cases}1 & (n=0)\\ 0 & (n\neq 0)\end{cases}$$



$$\delta(n-n_0)=\begin{cases}1 & (n=n_0)\\ 0 & (n\neq n_0)\end{cases}$$



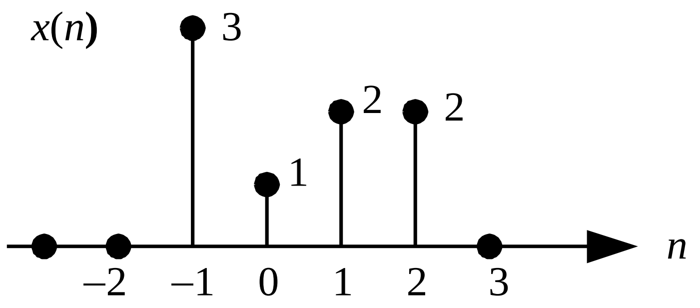
Note: $\delta(t)$ is represented by area (intensity)(as $t\to 0$, the amplitude is ∞);
$\delta(n)$ takes a finite amplitude of 1 at ***n***=0 (not area).

Representing Any Discrete-Time Signal Using Unit Sample Sequences

$$x(n)=\sum_{m=-\infty}^{\infty}x(m)\delta(n-m)$$

where $x(m)\delta(n-m)=\begin{cases}x(n) & m=n\\ 0 & m\neq n\end{cases}$

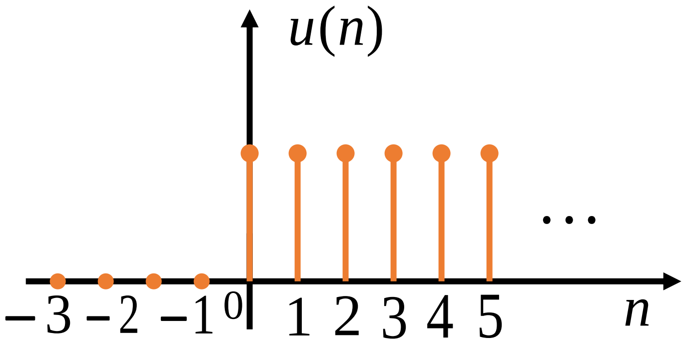
Any sequence is the sum of weighted and delayed unit sample sequences.



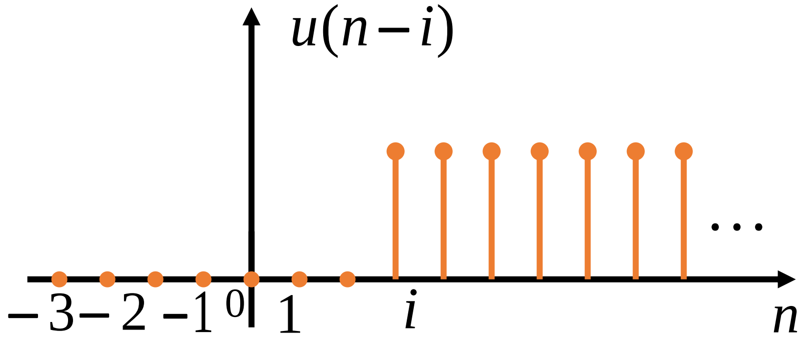
$$x(n)=3\delta(n+1)+\delta(n)+2\delta(n-1)+2\delta(n-2)$$

2. Unit Step Sequence

$$u(n)=\begin{cases}1 & (n\geq 0)\\ 0 & (n<0)\end{cases}$$



$$u(n-i)=\begin{cases}1 & (n\geq i)\\ 0 & (n<i)\end{cases}$$

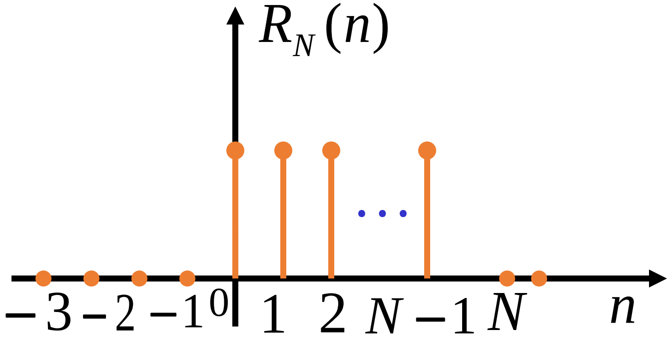


$$\begin{cases}u(n)=\sum_{K=0}^{\infty}\delta(n-K)\\ \delta(n)=u(n)-u(n-1)\end{cases}$$

$\delta(n)$ and $u(n)$ have a difference – sum relationship, not a calculus relationship anymore.

3. Rectangular Sequence

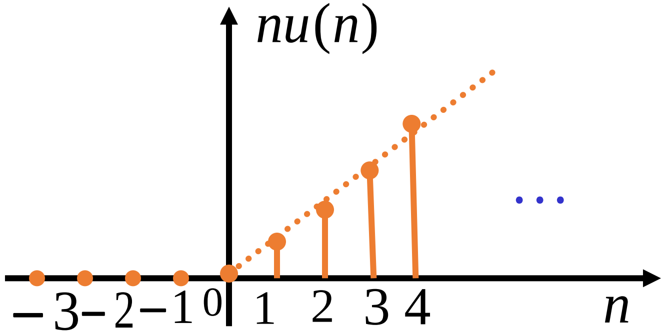
$$R_N(n) = \begin{cases} 1 & (0 \le n \le N-1) \\ 0 & (n < 0, n \ge N) \end{cases}$$



$$R_N(n) = u(n) - u(n-N) = \sum_{k=0}^{N-1} \delta(n-k)$$

4. Ramp Sequence

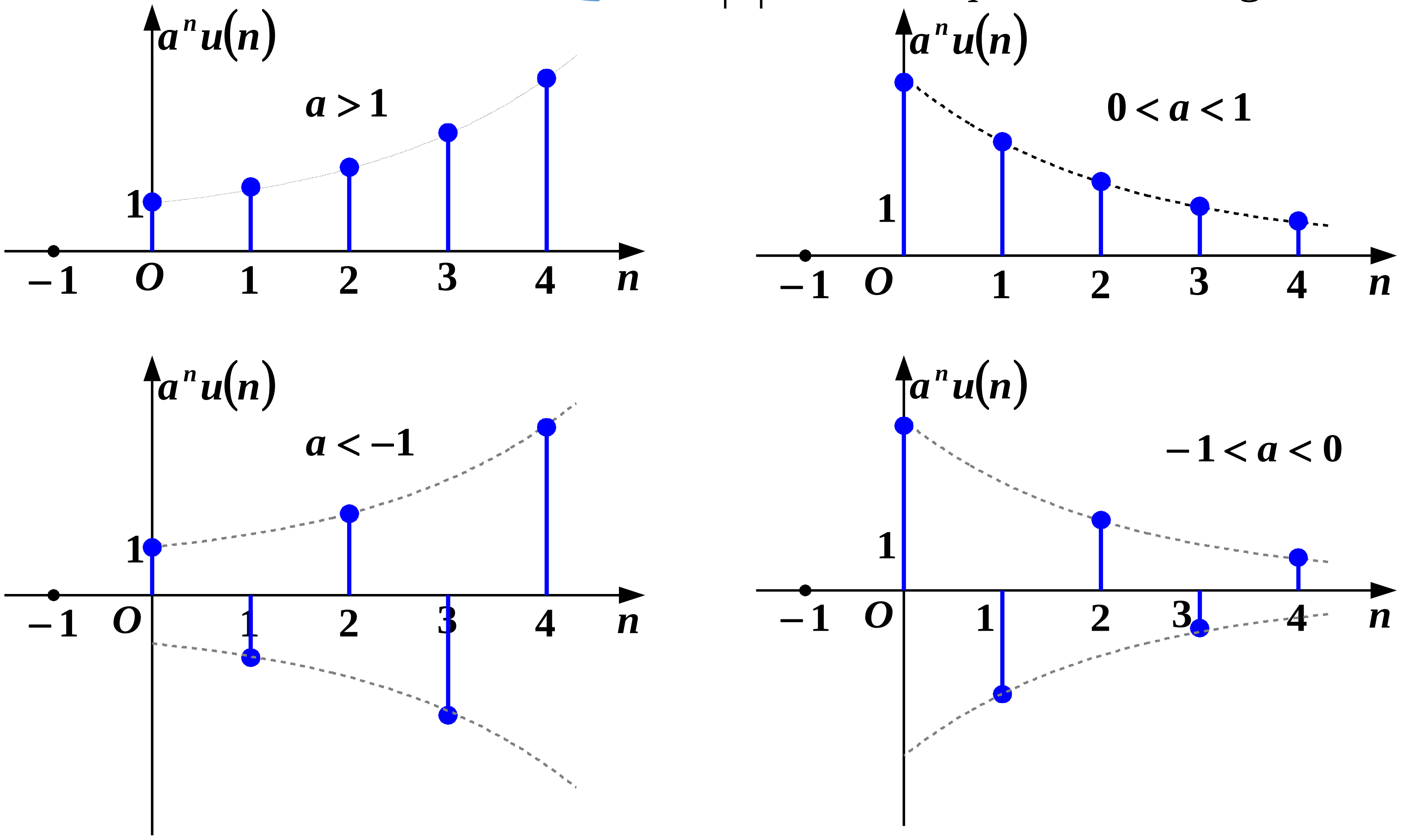
$$x(n) = nu(n)$$



5. Real-Valued Exponential Sequence

$$x(n) = a^n u(n)$$

$$\begin{cases} \text{when } |a| > 1 \text{ the sequence diverges;} \\ \text{when } |a| < 1 \text{ the sequence converges.} \end{cases}$$

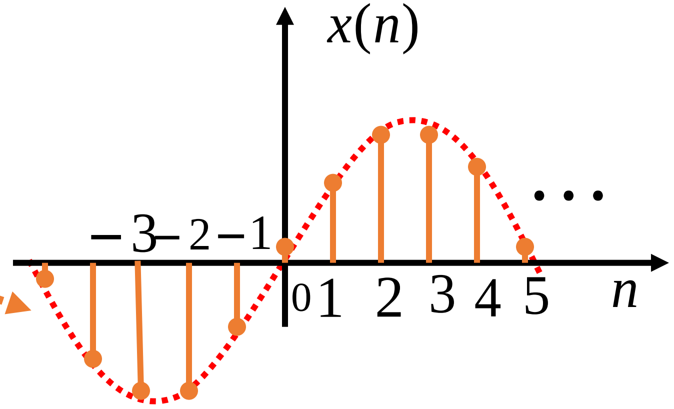


6. Sinusoidal Sequence

$$f(t) = \sin \Omega_0 t$$

$$x(n) = f(nT)$$

$$= \sin(\Omega_0 nT) = \sin(n\omega_0)$$



where ω_0 is the frequency of the sinusoidal sequence, reflecting the rate at which the sequence values repeat periodically in turn. For example, $\omega_0 = 2\pi / 10$.

$$\omega_0 = \Omega_0 T = \frac{\Omega_0}{f_s}$$

ω_0 is the normalized value of Ω_0 with respect to f_s

单选题 1分

设置

Is the sequence $\sin\frac{4\pi}{11}n$ a periodic signal? If yes, what is its period?

A No

B Yes， 11/4

C Yes， 11/2

D Yes， 11

提交

Determination of Periodicity of Sine Sequences:

① $\frac{2\pi}{\omega_0} = N$, N is a positive integer

$$\sin\left[\omega_0\left(n+N\right)\right] = \sin\left[\omega_0\left(n+\frac{2\pi}{\omega_0}\right)\right] = \sin\left(\omega_0 n + 2\pi\right) = \sin\left(\omega_0 n\right)$$

The sine sequence is periodic, **with period N**

② $\frac{2\pi}{\omega_0} = \frac{N}{m}$, $\frac{N}{m}$ is a rational number

$$\sin\left[\omega_0\left(n+N\right)\right] = \sin\left[\omega_0\left(n+m\frac{2\pi}{\omega_0}\right)\right] = \sin\left(\omega_0 n + m\cdot 2\pi\right) = \sin\left(\omega_0 n\right)$$

$\sin\left(\omega_0 n\right)$ is still periodic. **Period:** $m\frac{2\pi}{\omega_0} = N$

③ $\frac{2\pi}{\omega_0}$ is an irrational number

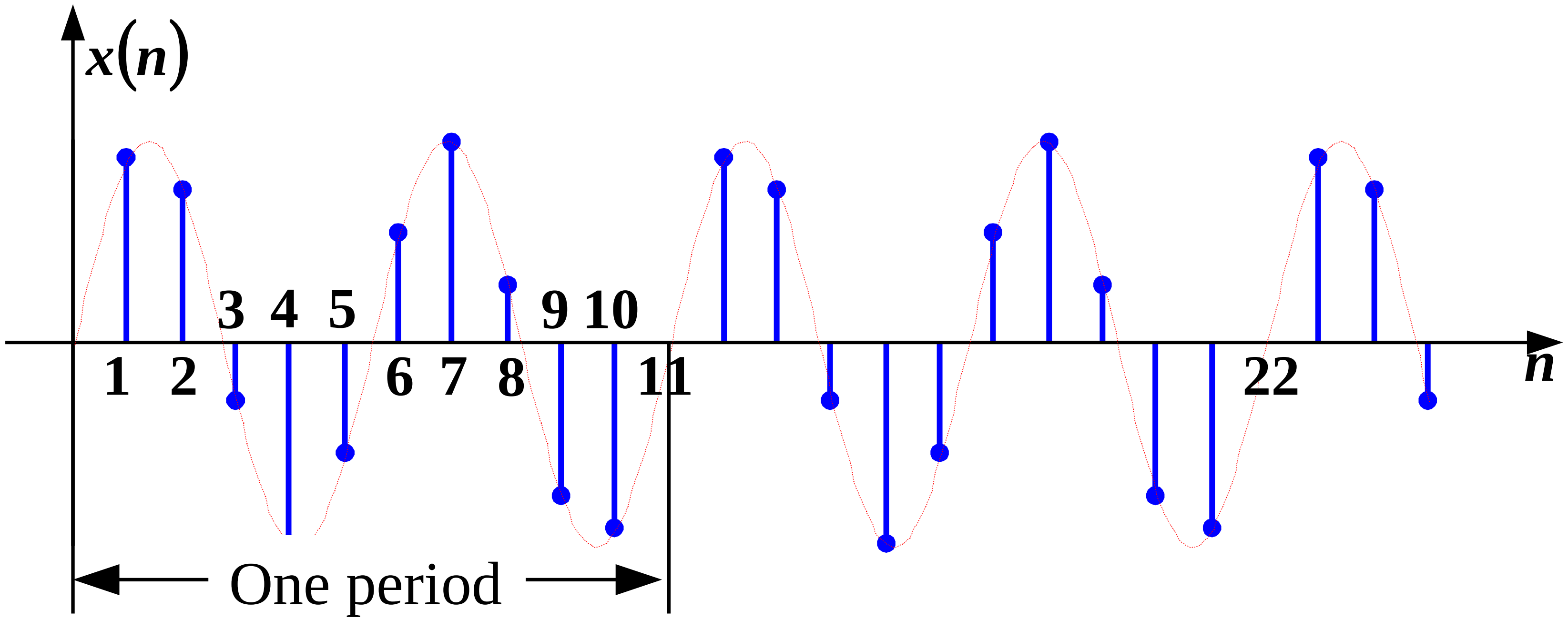
There is no value of N that satisfies $x\left(n+N\right) = x\left(n\right)$, so the sequence is **aperiodic**.

Example 6-1: Given $\sin\frac{4\pi}{11}n$, find its period.

Solution:

$$\omega_0=\frac{4\pi}{11}\text{ , } \textbf{then } \frac{2\pi}{\omega_0}=2\pi\frac{11}{4\pi}=\frac{11}{2}=\frac{N}{m}$$

so $N=11$, that is, the period is **11.** (There are 5.5 ω_0 in 2π)



7. Complex Exponential Sequence

$$x(n) = e^{j\omega_0 n} = \cos(\omega_0 n) + j\sin(\omega_0 n)$$

A complex sequence can be expressed in polar coordinates:

$$x(n) = |x(n)|e^{j\arg[x(n)]}$$

$$|x(n)| = 1 \qquad \arg[x(n)] = \omega_0 n$$

General complex exponential sequence

$$x(n) = Ae^{(\sigma + j\omega_0)n} = Ae^{\sigma n}e^{j\omega_0 n}$$

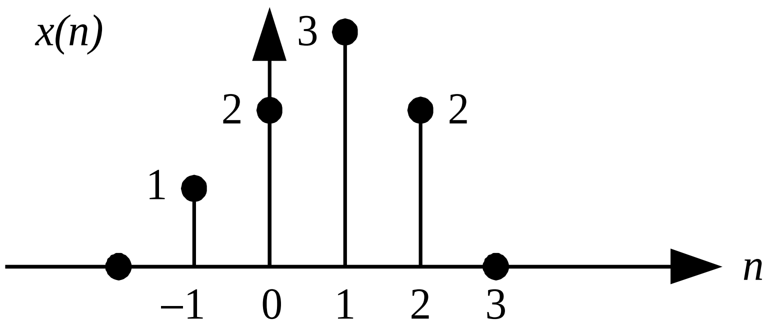
$$|x(n)| = Ae^{\sigma n} \qquad \arg[x(n)] = \omega_0 n$$

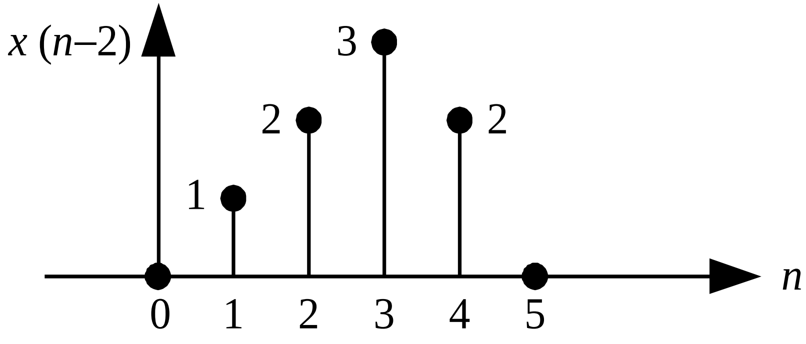
$$x(n) = Ae^{\sigma n}e^{j\omega_0 n} = Ae^{\sigma n}\cos(\omega_0 n) + jAe^{\sigma n}\sin(\omega_0 n)$$

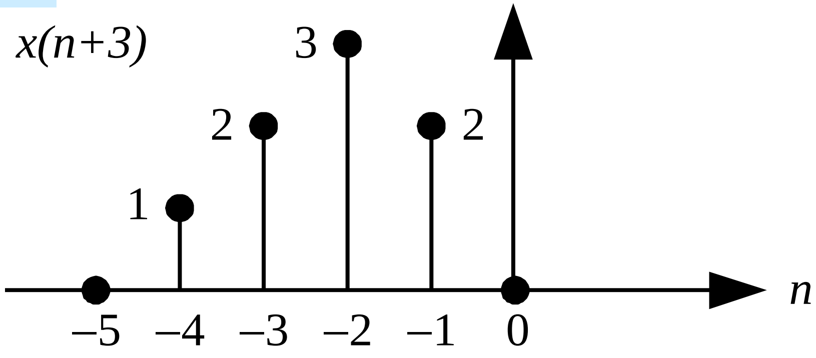
6.2.2 Operations on Discrete-Time Signals

1. Shifting of Sequences

$$\left.\begin{array}{l}\boldsymbol{x(n-m)} \text{ means shifting } \boldsymbol{x(n)} \text{ to the right by } \boldsymbol{m} \text{ units.} \\ \boldsymbol{x(n+m)} \text{ means shifting } \boldsymbol{x(n)} \text{ to the left by } \boldsymbol{m} \text{ units.}\end{array}\right\} m > 0$$

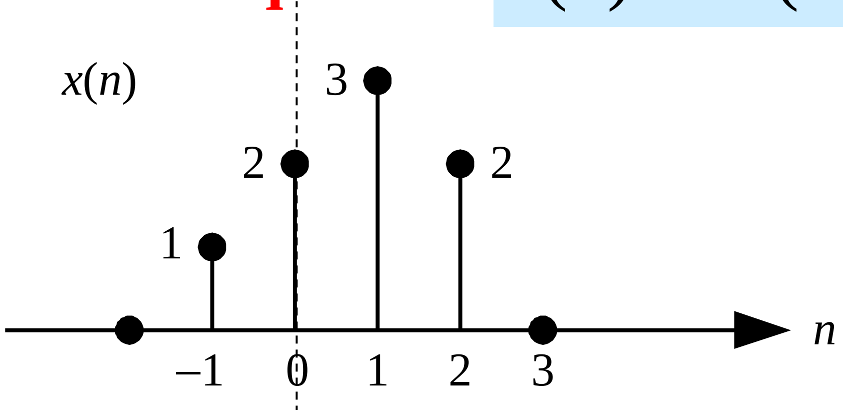


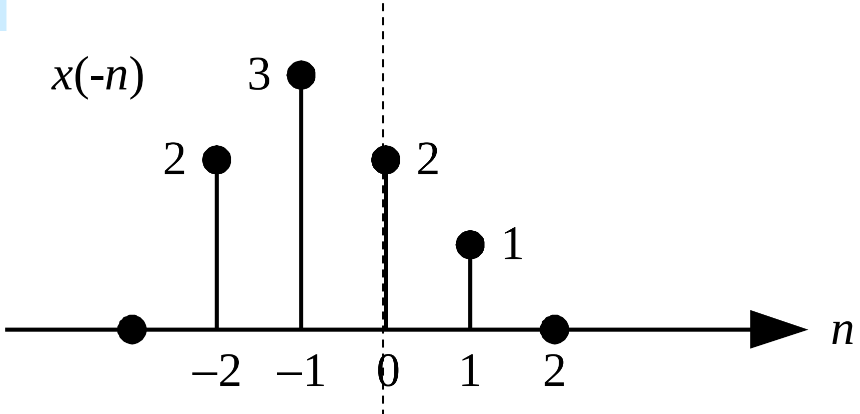




2. Reverse of Sequences

$$x(n) \to x(-n)$$





6.2 Discrete-Time Signals - Sequences

3. Scaling of Sequences

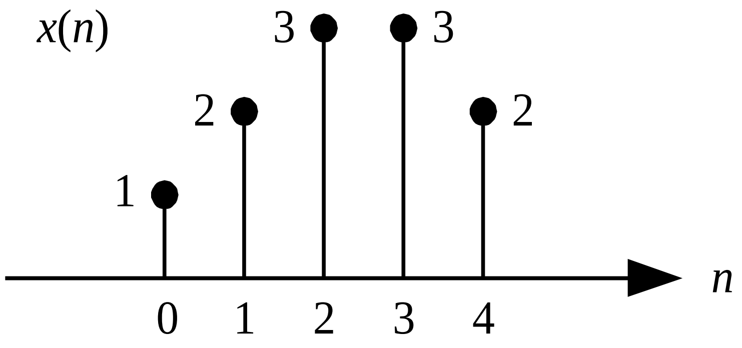
***a* > 0** is a positive integer.

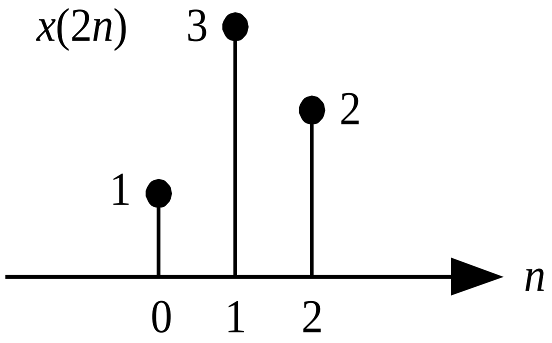
$$x(n) \rightarrow x(an)$$

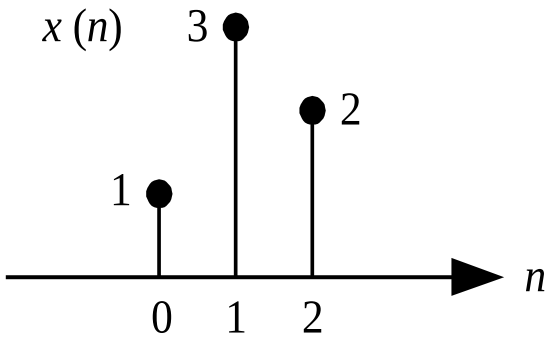
$$x(n) \rightarrow x\left(\frac{n}{a}\right)$$

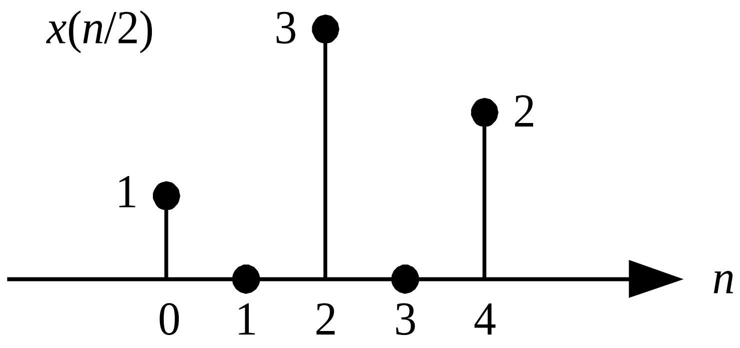
Compression: Extract one point every (*a*-1) points around the origin.

Expansion: Insert (*a*-1) zero-valued points between adjacent points.

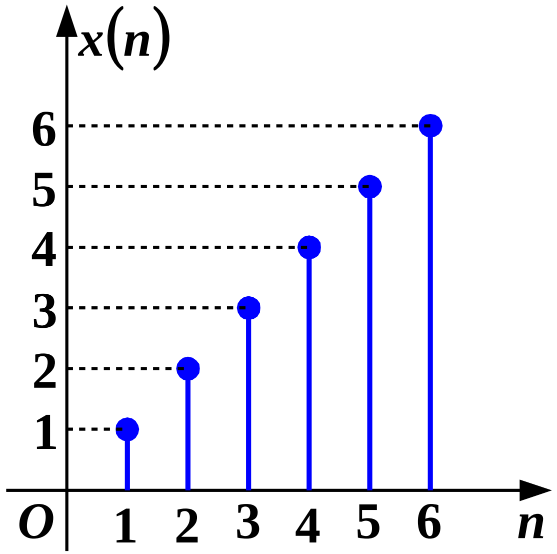




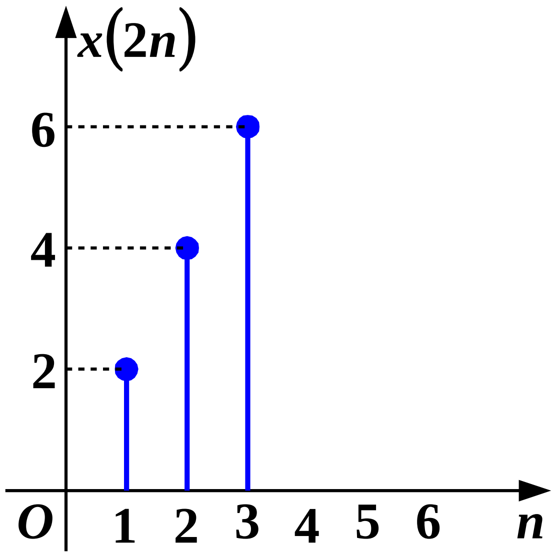


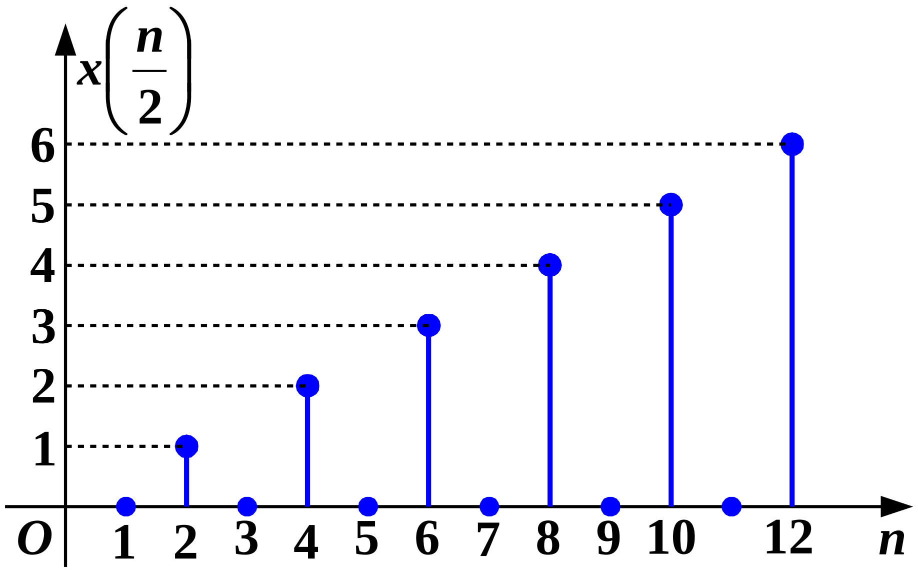


Example 6-2: Given the waveform of $x(n)$, please draw the waveforms of $x(2n)$ and $x(n/2)$.



Solution:

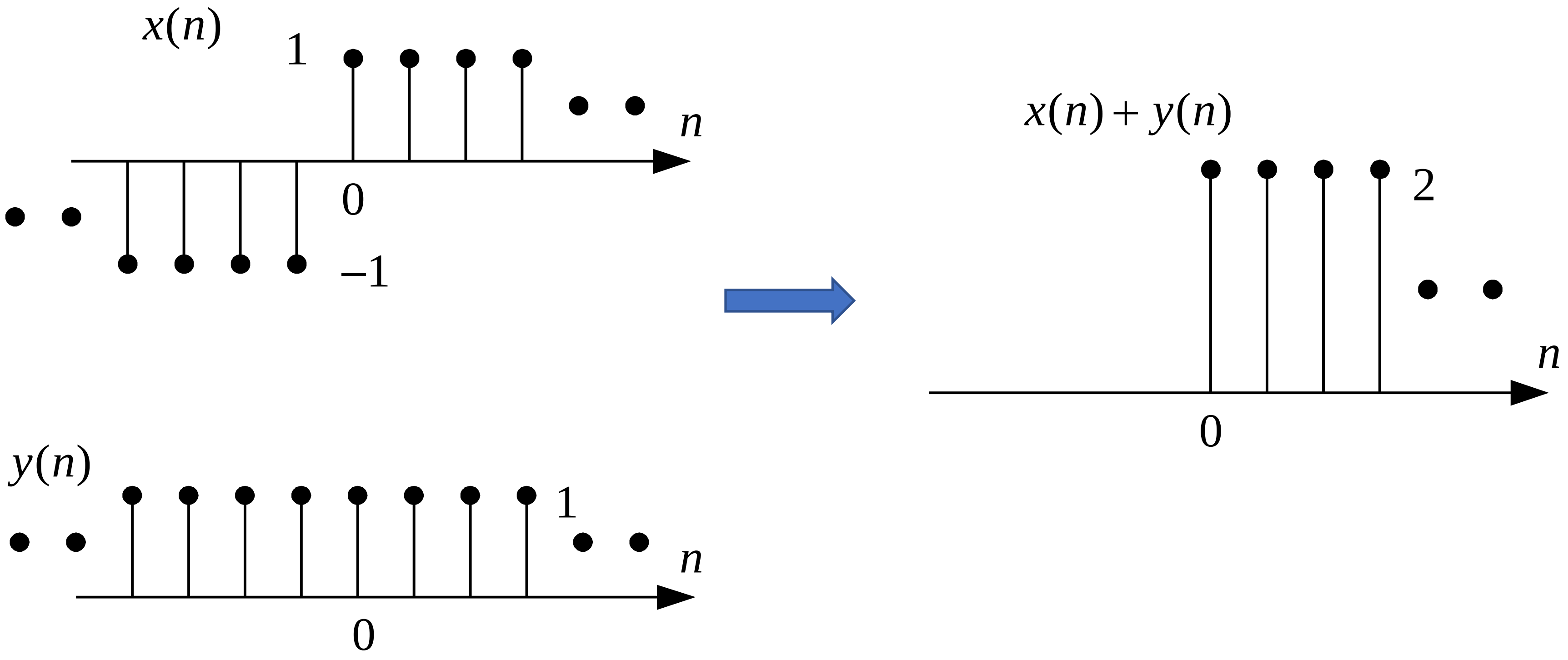




4. Addition of Sequences

$$z(n) = x(n) + y(n)$$

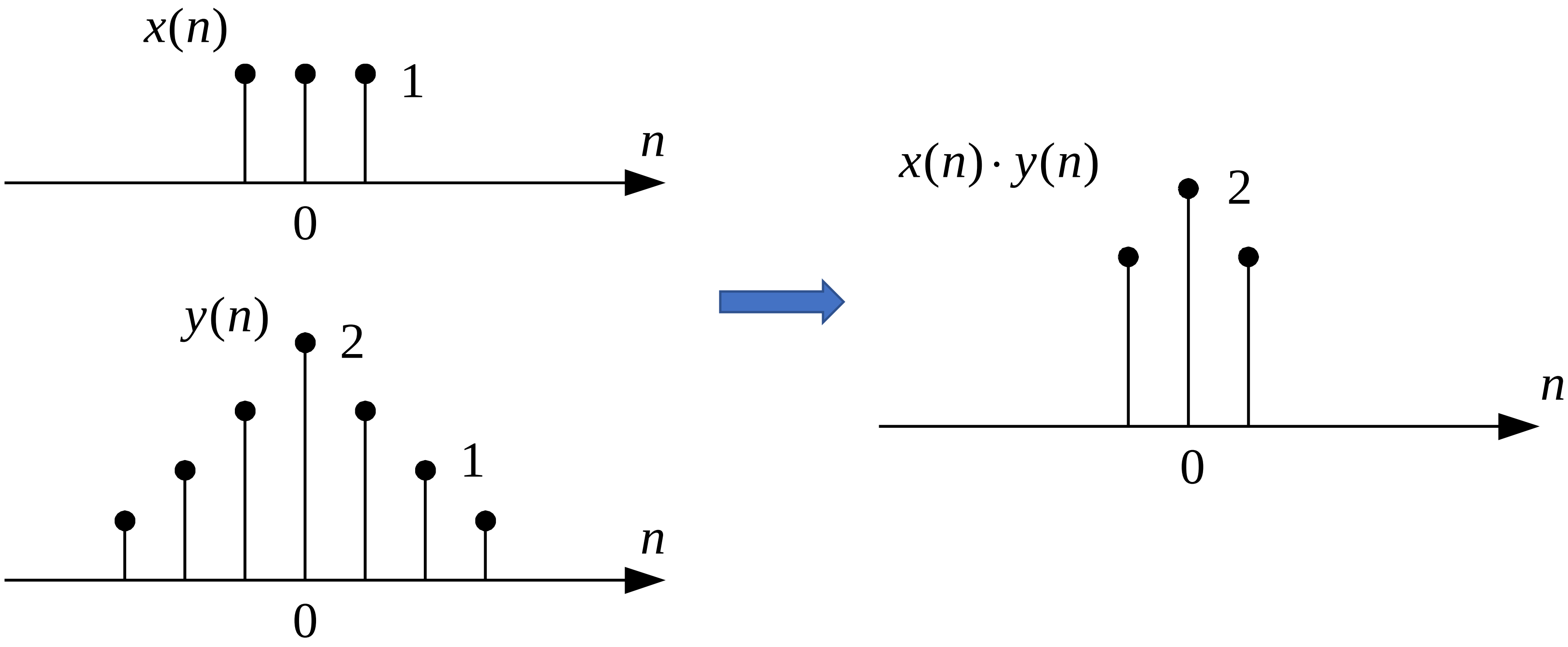
Add term by term.



5. Multiplication of Sequences

$$z(n) = x(n) \cdot y(n)$$

Multiply term by term.



6. Difference of Sequences

Backward Difference: Subtract the past value from the current value.

First-order backward difference $\nabla x(n) = x(n) - x(n-1)$

Second-order backward difference
$$\begin{aligned}\nabla^2 x(n) &= \nabla\{\nabla x(n)\}\\ &= [x(n)-x(n-1)]-[x(n-1)-x(n-2)]\\ &= x(n)-2x(n-1)+x(n-2)\end{aligned}$$

Forward Difference: Subtract the current value from the future value.

First-order forward difference $\Delta x(n) = x(n+1) - x(n)$

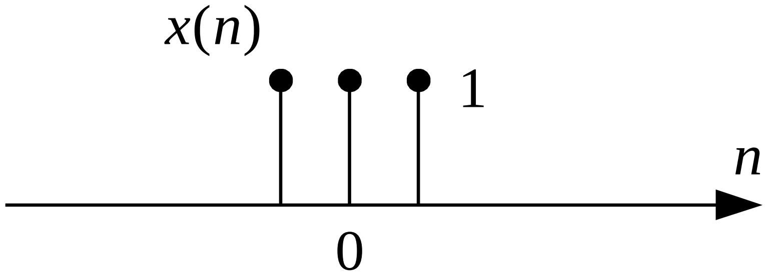
Second-order forward difference $\Delta^2 x(n) = \Delta\{\Delta x(n)\} = x(n+2) - 2x(n+1) + x(n)$

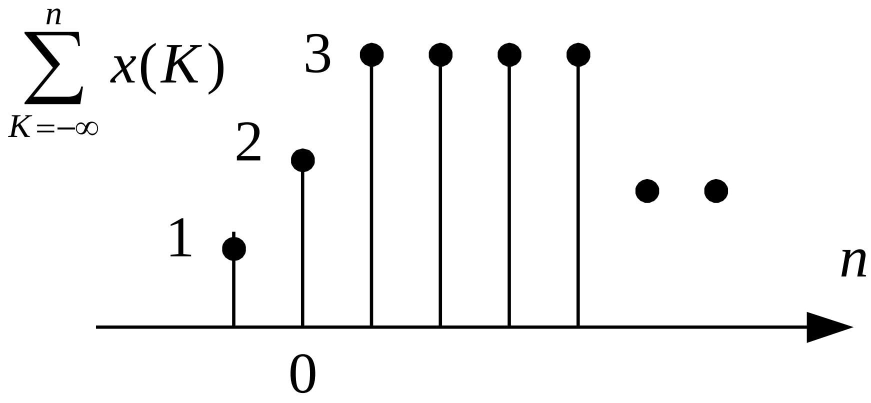
The unit sample sequence can be expressed by the difference of the unit step sequence:

$$\delta(n) = u(n) - u(n-1)$$

7. Summation of Sequences

$$z(n) = \sum_{K=-\infty}^{n} x(K)$$





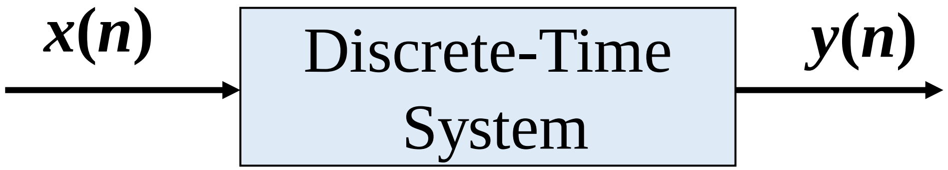
The unit step sequence can be expressed by the summation of the unit sample sequence:

$$u(n) = \sum_{K=-\infty}^{n} \delta(K)$$

8. Energy of a Sequence

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

6.3.1 Discrete Linear Time-Invariant System

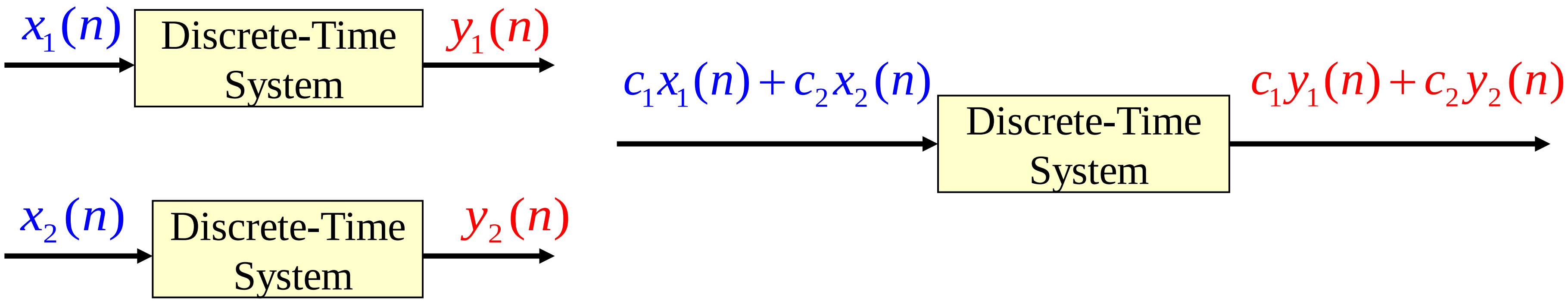


1. Linearity

Homogeneity and Superposition: Suppose two pairs of excitation and response

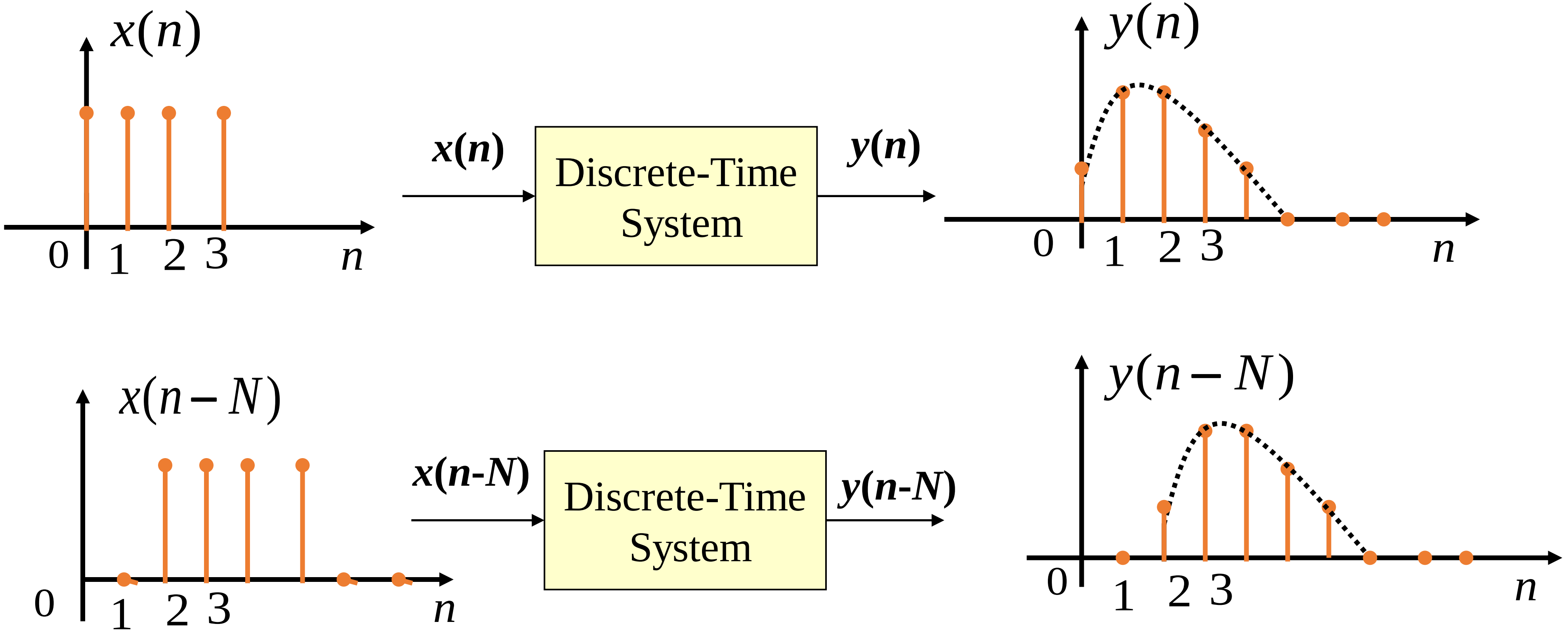
$$x_1(n) \to y_1(n),\ x_2(n) \to y_2(n)$$

$$c_1 x_1(n) + c_2 x_2(n) \to c_1 y_1(n) + c_2 y_2(n)$$



2. Time-Invariance

Suppose the excitation and response satisfy $x(n) \to y(n)$, then $x(n-N) \to y(n-N)$.



6.3.2 Mathematical Model of Discrete-Time Systems

Mathematical Model of Continuous Systems—Differential Equation

$$C_0 \frac{d^n r(t)}{dt^n} + C_1 \frac{d^{n-1} r(t)}{dt^{n-1}} + \ldots + C_n r(t) = E_0 \frac{d^m e(t)}{dt^m} + E_1 \frac{d^{m-1} e(t)}{dt^{m-1}} + \ldots + E_m e(t)$$

Basic Operations: Differentiation, Multiplication by Coefficients, Addition

Mathematical Model of Discrete Systems—Difference Equation

$$y(n) = -\sum_{k=1}^{N} a_k y(n-k) + \sum_{r=0}^{M} b_r x(n-r)$$

The input is the discrete sequence $x(n)$ and its time-shifted functions

$$x(n), x(n-1), x(n-2), \ldots.$$

The output is the discrete sequence $y(n)$ and its time-shifted functions

$$y(n), y(n-1), y(n-2), \ldots.$$

Basic Operations: Delay, Multiplication by Coefficients, Addition

6.3.3 Basic Operational Units of Discrete Systems

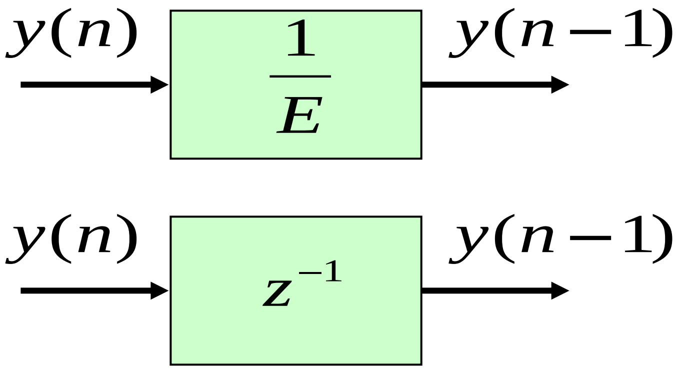
Delay Element (Unit Delay)

For LTI systems, concepts such as operator symbols and transfer operators can be used to represent or solve the mathematical model of the system. For discrete-time systems, the operator symbol E is used to denote the operation of advancing a sequence by one unit of time. E is also called a time-shift operator, and using the shift operator, we can write:

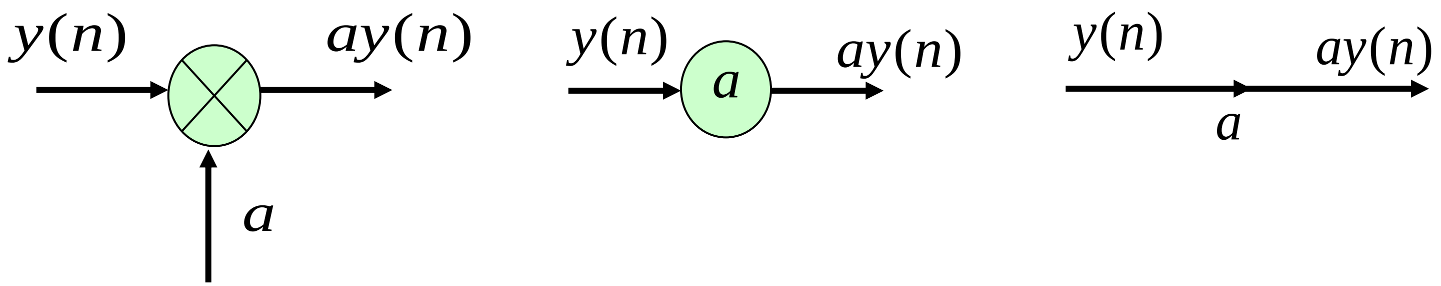
$$y(n+1) = Ey(n) \qquad y(n-1) = \frac{1}{E}y(n)$$

The above analysis shows that the operator $\frac{1}{E}$ represents the function of delaying by one unit of time, that is, $y(n)$ undergoes the $\frac{1}{E}$ operation to give $y(n-1)$. It is specified that $\frac{1}{E}$ is used as the symbol of the delay element.

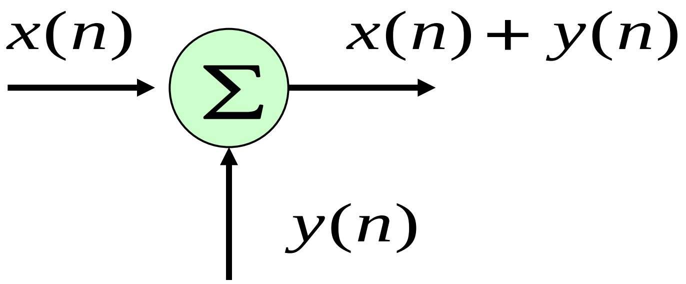
A unit delay is actually a shift register, which pushes out the previous discrete value and replenishes it.



Multiplier



Adder



6.3.5 Characteristics of Difference Equations

1. The *n*-th value of the output sequence is determined not only by the input sample at the same instant but also by the previous output values, and each output value must be retained sequentially.

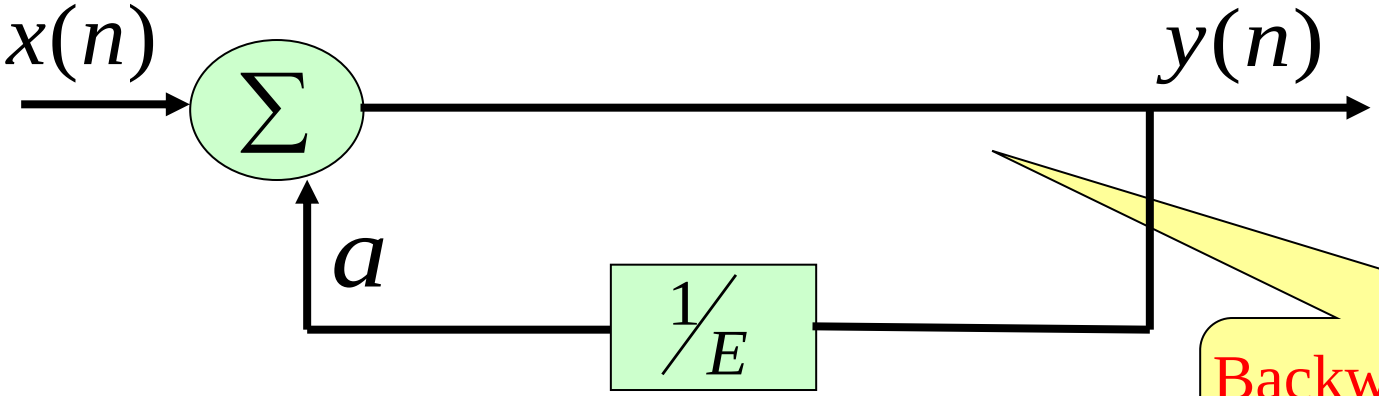
2. Order of Difference Equation: The order is the difference between the highest and lowest indices of variables in the difference equation. If the *n*-th output of a system is determined by several just-past output values and input values, the difference equation describing it is of that order.

General Form: $$\sum_{k=0}^{N} a_k y(n-k) = \sum_{r=0}^{M} b_r x(n-r)$$

3. Differential equations can be approximated by difference equations. The solution of a differential equation is an exact solution, while the solution of a difference equation is an approximate solution, and they have many similarities.

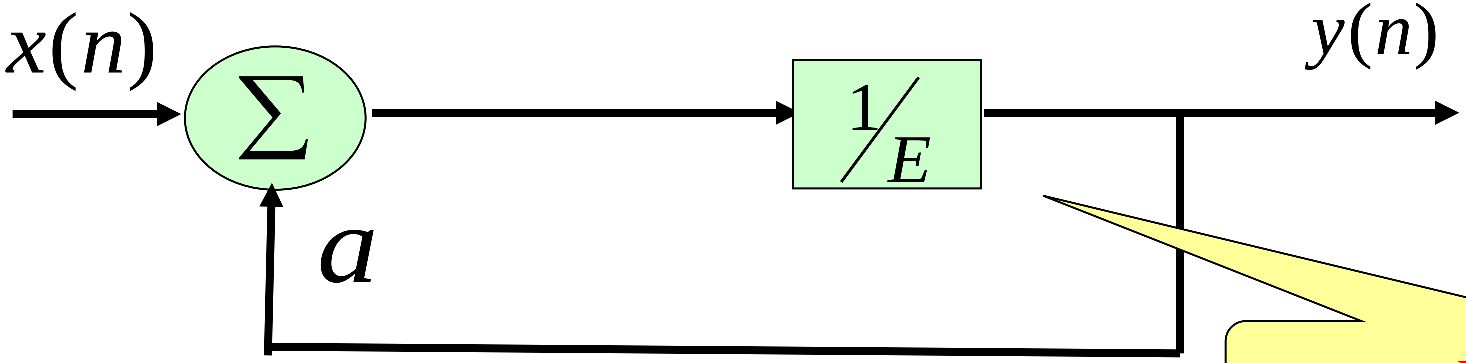
4. Difference equations describe discrete-time systems. The operational relationship between the input sequence and the output sequence corresponds to the system simulation block diagram, and one should be able to write and draw it.

There is no essential difference between the two systems !



Backward Difference Equation
(More Commonly Used)

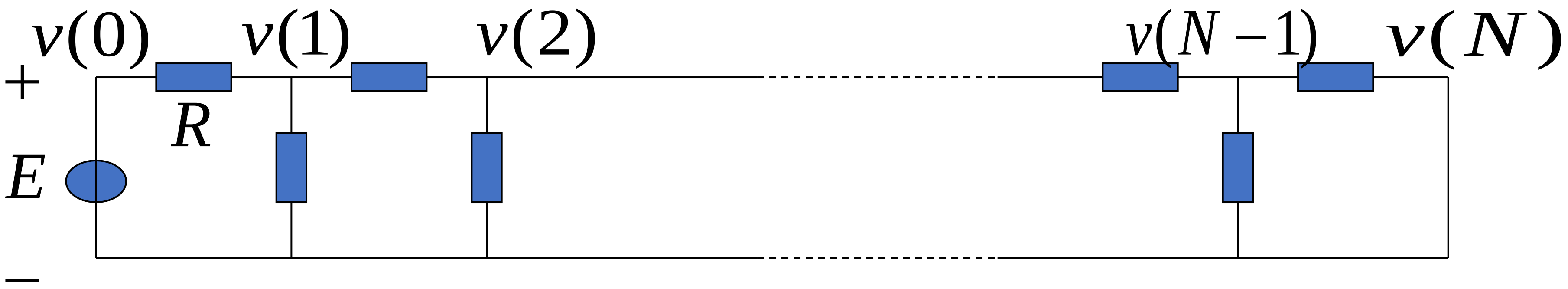
$$y(n) = x(n) + ay(n-1)$$



Forward Difference Equation

$$y(n+1) = ay(n) + x(n)$$

Example 6-3: A resistive ladder network, where each branch resistance is R, The voltage of each node to ground is $v(n)$, $n=0, 1, 2, \ldots, N$. Given the boundary node voltages $v(0)=E$, $v(N)=0$. derive the difference equation for $v(n)$.



Solution:

$$\frac{v(n-1)}{R}=\frac{v(n)-v(n-1)}{R}+\frac{v(n-2)-v(n-1)}{R}$$

$$v(n)-3v(n-1)+v(n-2)=0$$

Linear time-invariant discrete systems are described by linear difference equations with constant coefficients:

$$\sum_{k=0}^{N} a_k y(n-k) = \sum_{r=0}^{M} b_r x(n-r)$$

The order of the equation is equal to the difference between the highest and lowest indices of the unknown sequence variables.

Solving Methods:

- Iteration Method
- Classical Time-Domain Method: Find Homogeneous Solution and Particular Solution
- Convolution Method: Use the homogeneous solution to get the zero-input response, and then use the Convolution Sum to find the zero-state response.
- Transform Domain Method (z-Transform Method, Chapter 7)
- State Variable Analysis Method (Chapter 8)

The complete solution of a difference equation, i.e., the complete response of the system, consists of the homogeneous solution $y_h(n)$ and the particular solution $y_p(n)$:

$$y(n) = y_h(n) + y_p(n)$$

The form of the homogeneous solution $y_h(n)$ is determined by the characteristic roots of the homogeneous equation.

The form of the particular solution $y_p(n)$ is determined by the simplified form of the right-hand side of the equation after substituting the excitation signal.

1. Find the Homogeneous Solution (Inherent Response of the System)

Homogeneous equation of the difference equation:： $\sum_{k=0}^{N} a_k y(n-k) = 0$

$$a_0\alpha^N + a_1\alpha^{N-1} + \cdots + a_k\alpha^{N-k} + \cdots + a_N = 0$$

Characteristic Equation

The roots of the equation $\alpha_1, \alpha_2, \alpha_3 \ldots$

Characteristic Roots

α_i is called the natural frequency (or free frequency) of the system.

Homogeneous Solutions of Difference and Differential Equations

Characteristic Roots	Homogeneous Solution of Difference Equation	Homogeneous Solution of Differential Equation
Distinct Real Roots $\alpha_1, \alpha_2, \ldots, \alpha_N$	$\sum_{k=1}^{N} C_k \alpha_k^n$	$\sum_{k=1}^{N} A_k e^{\alpha_k t}$
k-fold Root α_1	$\sum_{i=1}^{k} C_i n^{k-i} \alpha_1^n$	$\sum_{i=1}^{k} A_i t^{k-i} e^{\alpha_1 t}$
Conjugate Complex Roots $\alpha \pm j\beta$	$\left(\sqrt{\alpha^2+\beta^2}\right)^n \left[P\cos(n\omega_0) + Q\sin(n\omega_0)\right]$ $\rho = \sqrt{\alpha^2+\beta^2} \quad P = C_1 + C_2 \quad Q = j(C_1 - C_2)$	$e^{\alpha t}(A\cos\beta t + B\sin\beta t)$

The undetermined coefficients of the homogeneous solution are determined by the boundary conditions (after finding the particular solution).

2. Find the Particular Solution: Determined by the Form of the Excitation Signal

Free Term (Right-Hand Side of the Equation)	Form of Particular Solution
α^n	$D\alpha^n$（if α is not a characteristic root） $Dn^r\alpha^n$（if α is an r-fold characteristic root）
n^k	$D_0 n^k + \cdots + D_{k-1}n + D_k$（if 1 is not a characteristic root） $n^r[D_0 n^k + \cdots + D_{k-1}n + D_k]$（if 1 is an r-fold characteristic root）
$\alpha^n n^k$	$[D_0 n^k + \cdots + D_{k-1}n + D_k]\alpha^n$（if α is not a characteristic root） $n^r[D_0 n^k + \cdots + D_{k-1}n + D_k]\alpha^n$（if α is an r-fold characteristic root）
$\cos(n\omega)$ or $\sin(n\omega)$	$D_1\cos(n\omega) + D_2\sin(n\omega)$（if $e^{\pm j\omega}$ is not a characteristic root）
$\alpha^n\cos(n\omega)$ or $\alpha^n\sin(n\omega)$	$\alpha^n[D_1\cos(n\omega) + D_2\sin(n\omega)]$ （if $\alpha e^{\pm j\omega}$ is not a characteristic root）

- By observing the function form of the free term on the right-hand side of the equation, try to select the form of the particular solution.
- Substitute the particular solution into the equation and use the method of undetermined coefficients to determine D_i。

3. Determine the Undetermined Coefficients of the Homogeneous Solution from Boundary Conditions

General Form of the Complete Solution (for Distinct Roots):

$$y(n) = C_1\alpha_1^{\ n} + C_2\alpha_2^{\ n} + \cdots + C_N\alpha_N^{\ n} + D(n) = \sum_{k=1}^{N} C_k\alpha_k^n + D(n)$$

C_i must be determined using N given boundary conditions. For example:

$$\begin{cases} y(0) = C_1 + C_2 + \cdots + C_N + D(0) \\ y(1) = C_1\alpha_1 + C_2\alpha_2 + \cdots + C_N\alpha_N + D(1) \\ \cdots\cdots \\ y(N-1) = C_1\alpha_1^{\ N-1} + C_2\alpha_2^{\ N-1} + \cdots + C_N\alpha_N^{\ N-1} + D(N-1) \end{cases}$$

For causal systems (where the excitation is applied at n=0), $y(-1), y(-2), \ldots, y(-N)$ are often given as boundary conditions.

Example 6-4: Find the complete solution of the difference equation

$$y(n)+2y(n-1)=x(n)-x(n-1)$$

The excitation signal is $x(n)=n^2$, and it is known that $y(-1)=-1$.

Solution: $\alpha=-2 \qquad \therefore \quad y_h(n)=C_1(-2)^n$

Characteristic root

Free term on the right-hand side $n^2-(n-1)^2=2n-1$

$$y_p(n)=D_0n+D_1$$

Form of Particular Solution

$$D_0n+D_1+2D_0(n-1)+2D_1=2n-1$$

Substitute the particular solution into the difference equation

$$\begin{cases} 3D_0=2 \\ 3D_1-2D_0=-1 \end{cases} \Rightarrow D_0=\frac{2}{3},\quad D_1=\frac{1}{9}$$

$$y_p(n)=\frac{2}{3}n+\frac{1}{9}$$

Particular Solution-Forced Response

Complete Solution = Homogeneous Solution + Particular Solution

$$y(n) = C_1(-2)^n + \frac{2}{3}n + \frac{1}{9}$$

Substitute the boundary condition to find the undetermined coefficient, C_1

$$y(-1) = C_1(-2)^{-1} + \frac{2}{3}(-1) + \frac{1}{9} = -1$$

$$C_1 = \frac{8}{9}$$

Obtain the closed-form of the complete solution:

$$y(n) = \frac{8}{9}(-2)^n + \frac{2}{3}n + \frac{1}{9}$$

Natural/Free Response

Forced Response

Homework

1. Repeat the examples in the lecture notes.

2. Determine whether the following sequence is a periodic signal. If yes, what is its period?

(1)$\cos\left(\frac{3}{7}n - \frac{\pi}{8}\right)$ (2)$e^{j\frac{\pi}{8}n}$

3. Given the difference equation below:

$$y(n) - \frac{5}{6}y(n-1) + \frac{1}{6}y(n-2) = x(n),$$

where $x(n) = u(n),\ y(-1) = 5, y(-2) = 1.$

(1) Solve for the total response $y(n)$.

(2) Determine the free response $y_h(n)$ and the forced response $y_p(n)$.

Review of Last Lecture

6.1 Introduction

6.2 Discrete-Time Signals – Sequences

6.3 Mathematical Model of Discrete-Time Systems

6.4 Classical Time-Domain Solution of Linear Constant-Coefficient Difference Equations

Continuous-Time System	**vs.**	**Discrete-Time System**
Differential Equation		Difference Equation
Convolution Integral		Convolution Sum
Laplace Transform		z-Transform
Fourier Transform		Discrete-Time Fourier Transform, Discrete Fourier Transform
System Function		System Function
Convolution Theorem		Convolution Theorem

6.2 Discrete-Time Signals - Sequences

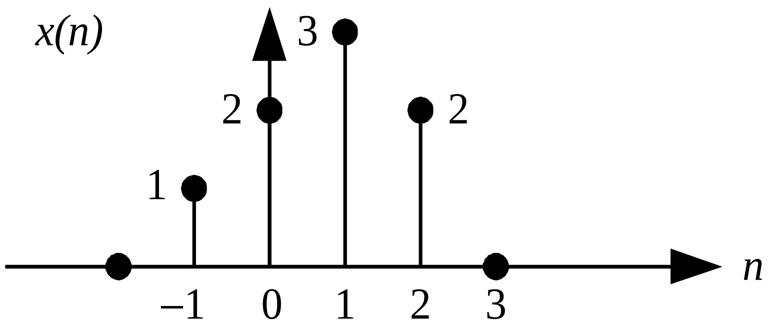
6.2.2 Operations on Discrete-Time Signals

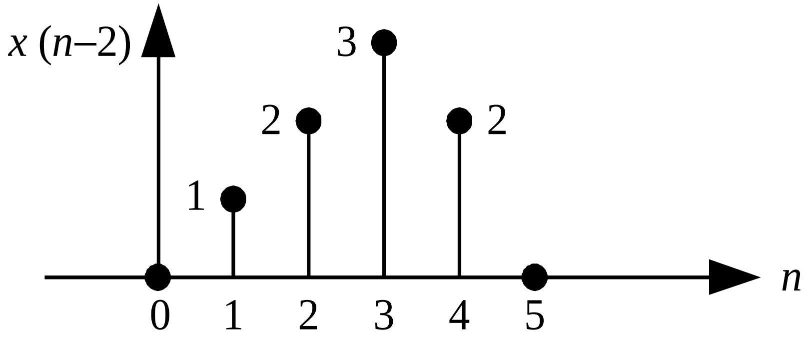
1. Shifting of Sequences

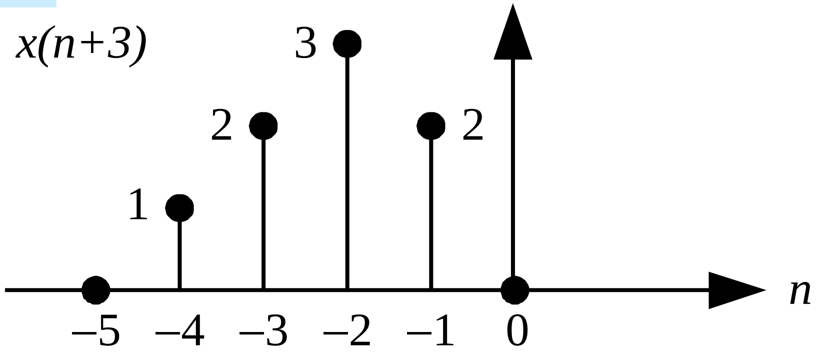
$x(n-m)$ means shifting **$x(n)$** to the right by **m** units.

$x(n+m)$ means shifting **$x(n)$** to the left by **m** units.

$m > 0$

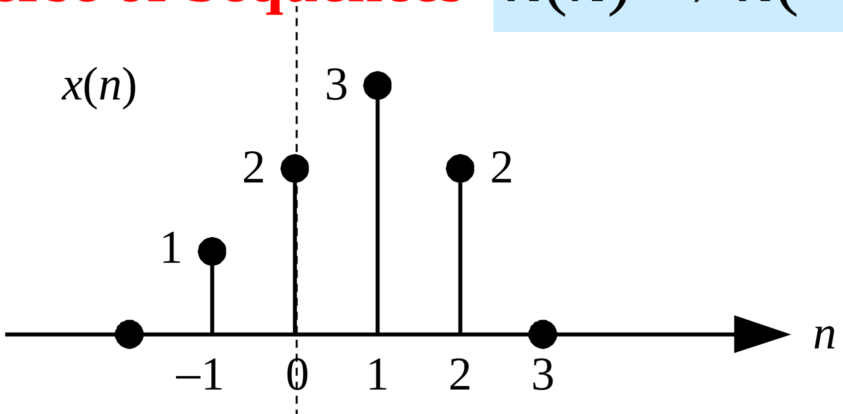


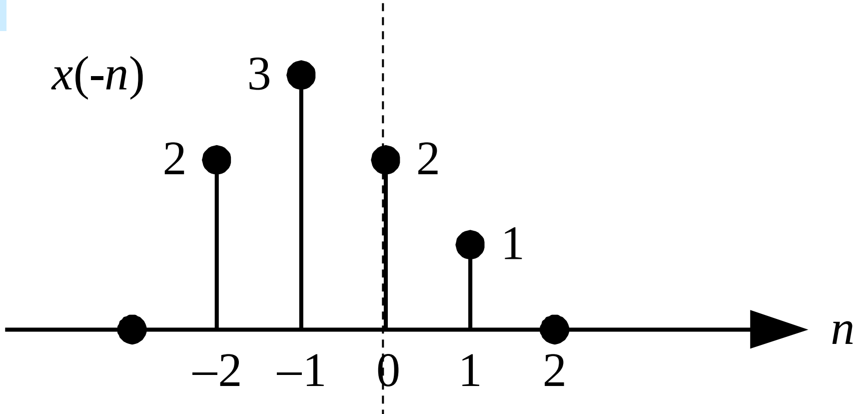




2. Reverse of Sequences

$$x(n) \to x(-n)$$





单选题 1分

设置

Given sequence $x(n)=\{-2,1,2,\overset{\downarrow}{0},3,2,4,6,5\}$, then $x(2n)=(\quad)$

A $\{-2,\overset{\downarrow}{2},3,4,5\}$

B $\{1,\overset{\downarrow}{0},2,6\}$

C $\{2,\overset{\downarrow}{0},3,2\}$

D $\{-2,1,\overset{\downarrow}{0},2,6\}$

提交

3. Scaling of Sequences

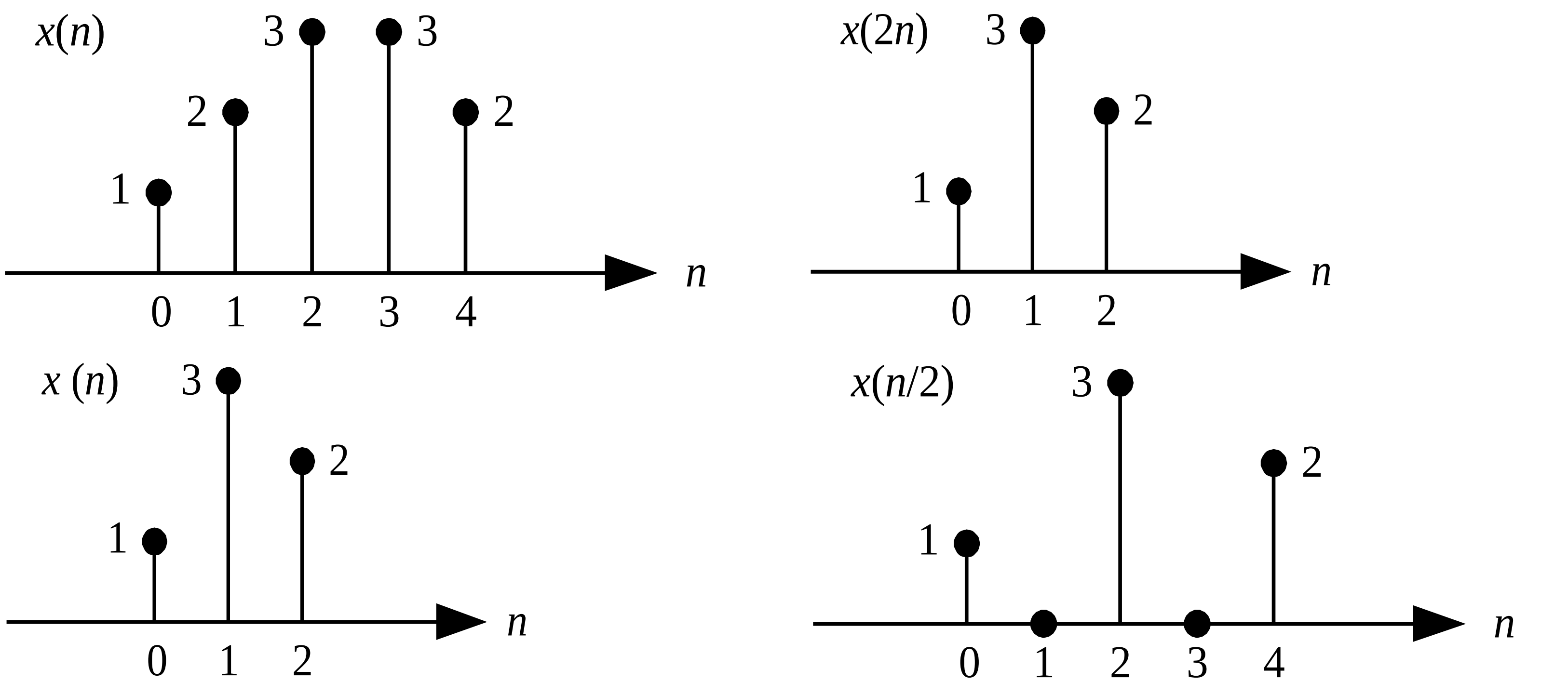
a **> 0** is a positive integer.

$$x(n) \to x(an)$$

$$x(n) \to x\left(\frac{n}{a}\right)$$

Compression: Extract one point every (*a*-1) points around the origin.

Expansion: Insert (*a*-1) zero-valued points between adjacent points.



Mathematical Model of Continuous Systems—Differential Equation

$$C_0 \frac{d^n r(t)}{dt^n} + C_1 \frac{d^{n-1} r(t)}{dt^{n-1}} + \ldots + C_n r(t) = E_0 \frac{d^m e(t)}{dt^m} + E_1 \frac{d^{m-1} e(t)}{dt^{m-1}} + \ldots + E_m e(t)$$

Basic Operations: Differentiation, Multiplication by Coefficients, Addition

Mathematical Model of Discrete Systems—Difference Equation

$$y(n) = -\sum_{k=1}^{N} a_k y(n-k) + \sum_{r=0}^{M} b_r x(n-r)$$

The input is the discrete sequence $x(n)$ and its time-shifted functions

$$x(n), x(n-1), x(n-2), \ldots.$$

The output is the discrete sequence $y(n)$ and its time-shifted functions

$$y(n), y(n-1), y(n-2), \ldots.$$

Basic Operations: Delay, Multiplication by Coefficients, Addition

The complete solution of a difference equation, i.e., the complete response of the system, consists of the homogeneous solution $y_h(n)$ and the particular solution $y_p(n)$:

$$y(n) = y_h(n) + y_p(n)$$

The form of the homogeneous solution $y_h(n)$ is determined by the characteristic roots of the homogeneous equation.

The form of the particular solution $y_p(n)$ is determined by the simplified form of the right-hand side of the equation after substituting the excitation signal.

1. Find the Homogeneous Solution (Inherent Response of the System)

Homogeneous equation of the difference equation:： $\sum_{k=0}^{N} a_k y(n-k) = 0$

$$a_0\alpha^N + a_1\alpha^{N-1} + \cdots + a_k\alpha^{N-k} + \cdots + a_N = 0$$

Characteristic Equation

The roots of the equation $\alpha_1, \alpha_2, \alpha_3 \dots$

Characteristic Roots

α_i is called the natural frequency (or free frequency) of the system.

Homogeneous Solutions of Difference and Differential Equations

Characteristic Roots	Homogeneous Solution of Difference Equation	Homogeneous Solution of Differential Equation
Distinct Real Roots $\alpha_1, \alpha_2, \ldots, \alpha_N$	$\sum_{k=1}^{N} C_k \alpha_k^n$	$\sum_{k=1}^{N} A_k e^{\alpha_k t}$
k-fold Root α_1	$\sum_{i=1}^{k} C_i n^{k-i} \alpha_1^n$	$\sum_{i=1}^{k} A_i t^{k-i} e^{\alpha_1 t}$
Conjugate Complex Roots $\alpha \pm j\beta$	$\left(\sqrt{\alpha^2+\beta^2}\right)^n \left[P\cos(n\omega_0)+Q\sin(n\omega_0)\right]$ $\rho=\sqrt{\alpha^2+\beta^2}$ $P=C_1+C_2$ $Q=j(C_1-C_2)$	$e^{\alpha t}(A\cos\beta t + B\sin\beta t)$

The undetermined coefficients of the homogeneous solution are determined by the boundary conditions (after finding the particular solution).

6.4 Classical Time-Domain Solution of Linear Difference Equations

2. Find the Particular Solution: Determined by the Form of the Excitation

Free Term (Right-Hand Side of the Equation)	Form of Particular Solution
α^n	$D\alpha^n$ (if α is not a characteristic root) $Dn^r\alpha^n$ (if α is an r-fold characteristic root)
n^k	$D_0n^k+\cdots+D_{k-1}n+D_k$ (if 1 is not a characteristic root) $n^r[D_0n^k+\cdots+D_{k-1}n+D_k]$ (if 1 is an r-fold characteristic root)
$\alpha^n n^k$	$[D_0n^k+\cdots+D_{k-1}n+D_k]\alpha^n$ (if α is not a characteristic root) $n^r[D_0n^k+\cdots+D_{k-1}n+D_k]\alpha^n$ (if α is an r-fold characteristic root)
$\cos(n\omega)$ or $\sin(n\omega)$	$D_1\cos(n\omega)+D_2\sin(n\omega)$ (if $e^{\pm j\omega}$ is not a characteristic root)
$\alpha^n\cos(n\omega)$ or $\alpha^n\sin(n\omega)$	$\alpha^n[D_1\cos(n\omega)+D_2\sin(n\omega)]$ (if $\alpha e^{\pm j\omega}$ is not a characteristic root)

- By observing the function form of the free term on the right-hand side of the equation, try to select the form of the particular solution.
- Substitute the particular solution into the equation and use the method of undetermined coefficients to determine D_i。

3. Determine the Undetermined Coefficients of the Homogeneous Solution from the Boundary Conditions

General Form of the Complete Solution (for Distinct Roots):

$$y(n) = C_1\alpha_1^{\ n} + C_2\alpha_2^{\ n} + \cdots + C_N\alpha_N^{\ n} + D(n) = \sum_{k=1}^{N} C_k\alpha_k^n + D(n)$$

C_i must be determined using N given boundary conditions. For example:

$$\begin{cases} y(0) = C_1 + C_2 + \cdots + C_N + D(0) \\ y(1) = C_1\alpha_1 + C_2\alpha_2 + \cdots + C_N\alpha_N + D(1) \\ \cdots\cdots \\ y(N-1) = C_1\alpha_1^{\ N-1} + C_2\alpha_2^{\ N-1} + \cdots + C_N\alpha_N^{\ N-1} + D(N-1) \end{cases}$$

For causal systems (where the excitation is applied at n=0), $y(-1), y(-2), \ldots, y(-N)$ are often given as boundary conditions.

Disadvantages of the Classical Method:

1. If the excitation term on the right side of the difference equation is relatively complex, it is difficult to handle.

2. If the excitation changes, the entire solution process must be repeated.

3. If the initial conditions change, the entire solution process must be repeated.

4. This method is a purely mathematical approach and cannot highlight the physical concepts of system response.

$$y(n)=\sum_{k=1}^{N}C_k\alpha_k^n+D(n)$$

Contents of This Lecture

Learning Objectives of This Lecture

1. Familiarize yourself with the classical time-domain solution method for difference equations and use **boundary conditions** correctly;
2. Familiarize yourself with the solution of **the unit sample response**;
3. Master the use of **convolution sum** to find the **zero-state response** proficiently.

Complete Solution = Homogeneous Solution + Particular Solution

Total Response = Natural Response + Forced Response

Total Response = Zero-Input Response + Zero-State Response

$$y(n) = \underbrace{\sum_{k=1}^{N} C_k \alpha_k^n}_{\text{Natural Response}} + \underbrace{D(n)}_{\text{Forced Response}}$$

$$= \underbrace{\sum_{k=1}^{N} C_{zik} \alpha_k^n}_{\text{Zero-Input Response}} + \underbrace{\sum_{k=1}^{N} C_{zsk} \alpha_k^n + D(n)}_{\text{Zero-State Response}}$$

1. Zero-Input Response

- **Definition of Zero-Input Response**: The response generated solely by the initial state of the system (with the input signal being zero), denoted as $\boldsymbol{y_{zi}(n)}$.

- Mathematical Model:

$$\sum_{k=0}^{N} a_k y(n-k) = 0$$

- Solution Method: Determine the form of the zero-input response based on the characteristic roots of the difference equation, then solve for the **undetermined coefficients of the homogeneous solution** using **boundary conditions**.

2. Zero-State Response

➢**Definition of Zero-State Response**: The zero-state response generated by the excitation $\boldsymbol{x(n)}$ when the initial state of the system is zero, denoted as $\boldsymbol{y_{zs}(n)}$.

➢Mathematical Model:
$$\sum_{k=0}^{N} a_k y(n-k) = \sum_{r=0}^{M} b_r x(n-r)$$

➢Solution :
$$y_{zs}(n) = \sum_{k=1}^{N} C_{zsk} \alpha_k^n + D(n)$$

For causal systems (the excitation signal is applied at **n=0, y(-1), y(-2), …,y(-N)** are usually given as boundary conditions.

The zero state means y(-1),y(-2),y(-3),…,y(-N) are all zero.

y(0),y(1),y(2),…,y(N-1) can be obtained by the iterative method.

For the N-th-order difference equation, to determine the coefficients of the homogeneous solution for the **zero-input response** or **zero-state response,** boundary conditions (i.e., N values of $y(n)$) are required.

➢ Determining coefficients of the homogeneous solution for **zero-input response**:

The n in the initial condition $y(n)$ can be either **positive or negative** and is **not necessarily continuous**. $y(-1), y(-2),\ldots y(-N)$ **or** $y(0), y(1),\ldots y(N-1)$ can all be used as **boundary conditions.** For $y(n)$ when $n\geq 0$, it can be obtained by iteratively substituting $y(n)$ ($n<0$) into the difference equation, and **the right-hand side of the difference equation must be set to zero during iteration**.

➢ Determining coefficients of the homogeneous solution for **zero-state response**:

Let $y(-1)=y(-2)=\ldots=y(-N+1)=0$, and iteratively find $y(0), y(1), \ldots$. **Assuming the excitation is applied at $n = 0$, the boundary conditions must include at least one $y(n)$ when $n\geq 0$, and n must be continuous when $n < 0$** (to ensure $y(n)$ with $n\geq 0$ can be obtained by iteration) . For example, $y(0)$, $y(-1)$, $\ldots y(-N+1)$ **can be used as boundary conditions**.

Example 6-5: Given the input signal $x(n)=u(n)$ and the difference equation of a discrete-time LTI system:

$$y(n)-0.9y(n-1)=0.05x(n)$$

(1) Find the zero-state response of the system.
(2) If the boundary condition **$y(-1)=1$,** find the zero-input response and the total response.

Solution:

(1) The homogeneous solution of the equation is $C(0.9)^n$.

According to the form of the excitation on the right-hand side of the equation, the particular solution is chosen as ***D.***

Substitute the particular solution into the difference equation: $D-0.9D=0.05$, $D=0.5$

The form of the complete solution is: $y(n)=C(0.9)^n+0.5$

From **$y(-1)=0$**, use the **iterative method** to get **$y(0) = 0.9y(-1)+0.05 = 0.05$**

Substitute $y(0)$ into $y(n)$: $0.05=C+0.5 \quad \Rightarrow C=-0.45$

$$\therefore\ y_{zs}(n)=\left[-0.45\times(0.9)^n+0.5\right]u(n)$$

(2) Find the **zero-input response**:

Set the excitation to zero, and the difference equation is:

$$y(n)-0.9y(n-1)=0$$

The zero-input response is $y_{zi}(n)=C_{zi}\times(0.9)^n$

Substitute ***y*(-1) =1:** $1=C_{zi}\times(0.9)^{-1} \quad \Rightarrow C_{zi}=0.9$

Total response: $y(n)=y_{zs}(n)+y_{zi}(n)$

$$=\underbrace{\left[-0.45\times(0.9)^n+0.5\right]}_{\textbf{zero-state response}}+\underbrace{0.9\times(0.9)^n}_{\textbf{zero-input response}}$$

$$=\underbrace{0.45\times(0.9)^n}_{\textbf{free response}}+\underbrace{0.5}_{\textbf{forced response}} \qquad (n\geq 0)$$

Finding the Complete Response by the Classical Method

The homogeneous solution of the equation is $C(0.9)^n$

According to the form of the excitation on the right-hand side of the equation, the particular solution is chosen as ***D*.**

Substitute the particular solution into the difference equation:

$$D - 0.9D = 0.05\,, \quad D = 0.5$$

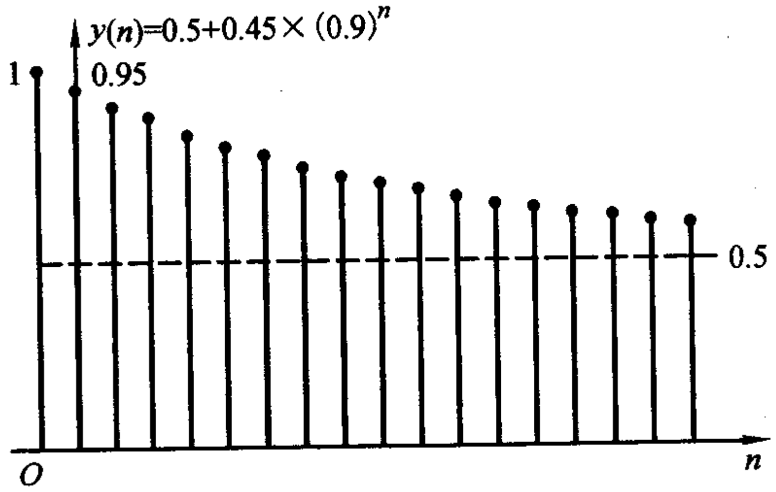
The form of the complete solution is: $y(n) = C(0.9)^n + 0.5$

From y(-1)=1, substituting into the equation gives y(0) = 0.9y(-1)+0.05 = 0.95

Substitute ***y*(0)=0.95** into the complete solution ***y*(*n*)** :

$$0.95 = C + 0.5 \quad \Rightarrow C = 0.45$$

$$\therefore\ y(n) = \left[0.45 \times (0.9)^n + 0.5\right] u(n)$$



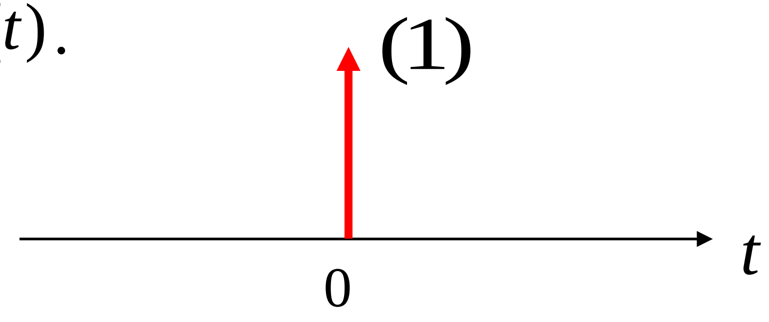
The result is the same as before!

6.6 Unit Sample Response of Discrete-Time Systems

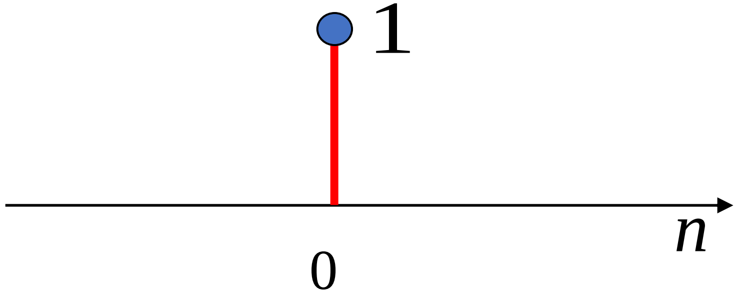
Unit Sample Response: It is the **zero-state response** of the system when the excitation is $\delta(n)$, denoted by ***h*(*n*)**.

Note the difference between $\delta(n)$ and $\delta(t)$.

$$\begin{cases} \int_{-\infty}^{\infty} \delta(t)dt = 1 & (t = 0) \\ \delta(t) = 0 & (t \neq 0) \end{cases}$$



$$\delta(n) = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$



Solution of Unit Sample Response: Convert the unit sample excitation signal into boundary conditions.

Convert $\delta(n)$ into boundary conditions, so finding the unit sample response is converted into finding **the homogeneous solution**, i.e., **the zero-input response when *n*>0.**

Example 6-6: Given the system's difference equation, find the unit sample response.

$$y(n)-3y(n-1)+2y(n-2)=x(n)$$

Solution: ***h*(*n*)** satisfies the equation

$$h(n)-3h(n-1)+2h(n-2)=\delta(n)$$

(1) Find Zero-Input Response

$$\alpha^2-3\alpha+2=0,\ \alpha_1=1\ \alpha_2=2$$

The homogeneous solution is: $C_1+C_2 2^n$

(2) Boundary Conditions

For a causal system, $h(-1)=0, h(-2)=0$

By substituting into the difference equation, we can derive:

$$h(0)=3h(-1)-2h(-2)+\delta(0)=1$$

$$h(1)=3h(0)-2h(-1)+\delta(1)=3$$

$$C_1 + C_2 2^n$$

Take $h(0) = 1, h(1) = 3$ as the boundary conditions for **the zero-input response**

$$\begin{cases} C_1 + C_2 = 1 \\ C_1 + 2C_2 = 3 \end{cases} \Rightarrow C_1 = -1 \quad C_2 = 2$$

(3) Find Unit Sample Response

$$h(n) = (-1 + 2 \cdot 2n)u(n) = (2^{n+1} - 1)u(n)$$

Note: The basic principle for choosing boundary conditions is that **the effect of $\delta(n)$ must be reflected in the boundary conditions**.

Analyzing Causality and Stability of Systems Based on Unit Sample Response

Causal System: A system where the output change does not precede the input change. That is, the response depends only on the current and previous excitations.

The **necessary and sufficient condition** for a discrete-time linear time-invariant system to be **a causal system** is:

$$h(n) = 0 \ (n < 0) \text{ or } \ h(n) = h(n)u(n)$$

Stable System: If the input is bounded, the output must be bounded.

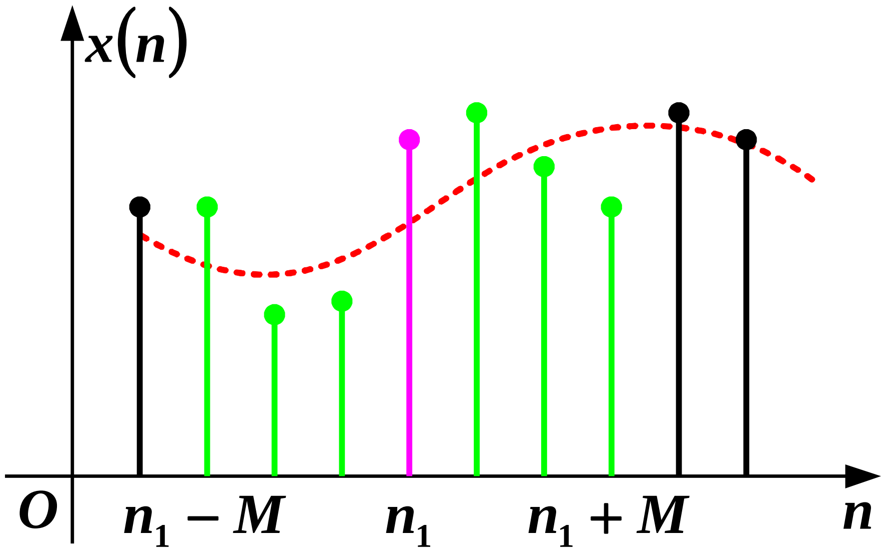
The **necessary and sufficient condition** for a discrete practical linear time-invariant system to be **a stable system** is:

$$\sum_{n=-\infty}^{\infty} |h(n)| \le M, M \text{ is a bounded positive vaule}$$

Example of a Non-causal System

Moving Average Filter: To observe the slow change trend over a period of time, a moving smoothing system can be used to filter out high-frequency components.

$$y(n)=\frac{1}{2M+1}\sum_{k=-M}^{M}x(n-k)$$



单选题 1分

设置

Given that the unit sample response of a system is $h(n) = a^n u(n)$, the system is ( ).

A Causal and stable

B Non-causal and stable

C Causal; stable when $|a| < 1$, unstable when $|a| > 1$

D Causal; unstable when $|a| < 1$, stable when $|a| > 1$

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Example 6-7： Given the unit sample response of a system $h(n)=a^n u(n)$, determine its causality and stability.

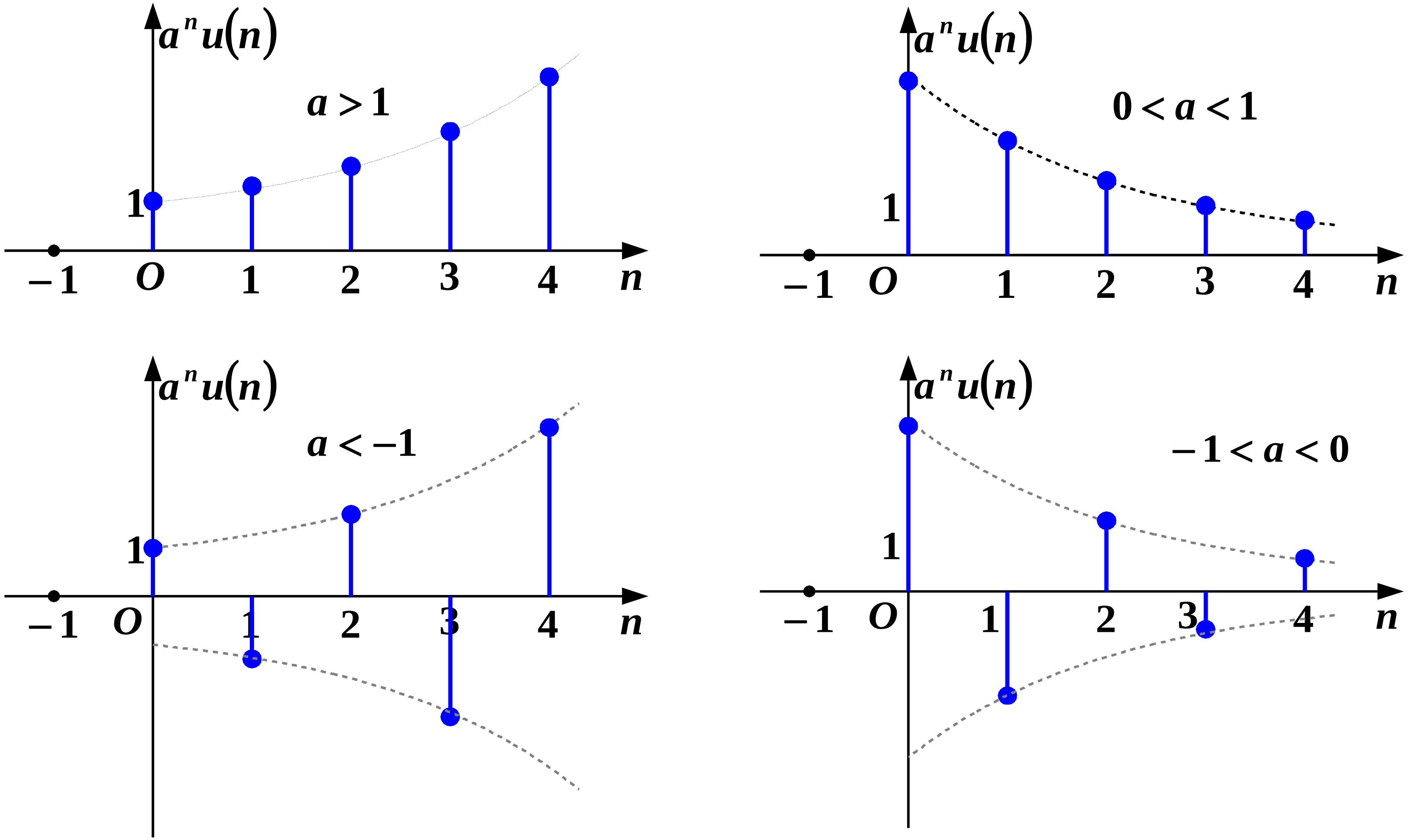
Solution： $n<0, u(n)=0 \quad \therefore n<0, h(n)=a^n u(n)=0$

Thus, the system is a **causal system**.

$$\sum_{n=-\infty}^{\infty}|h(n)|=\sum_{n=0}^{\infty}|a|^n=\begin{cases} |a|<1 & \dfrac{1}{1-|a|} & \text{bounded and stable} \\ |a|>1 & \lim\limits_{n\to\infty}\dfrac{1-|a|^{n+1}}{1-|a|} & \text{divergent and unstable} \end{cases}$$

6.6 Unit Sample Response of Discrete-Time Systems

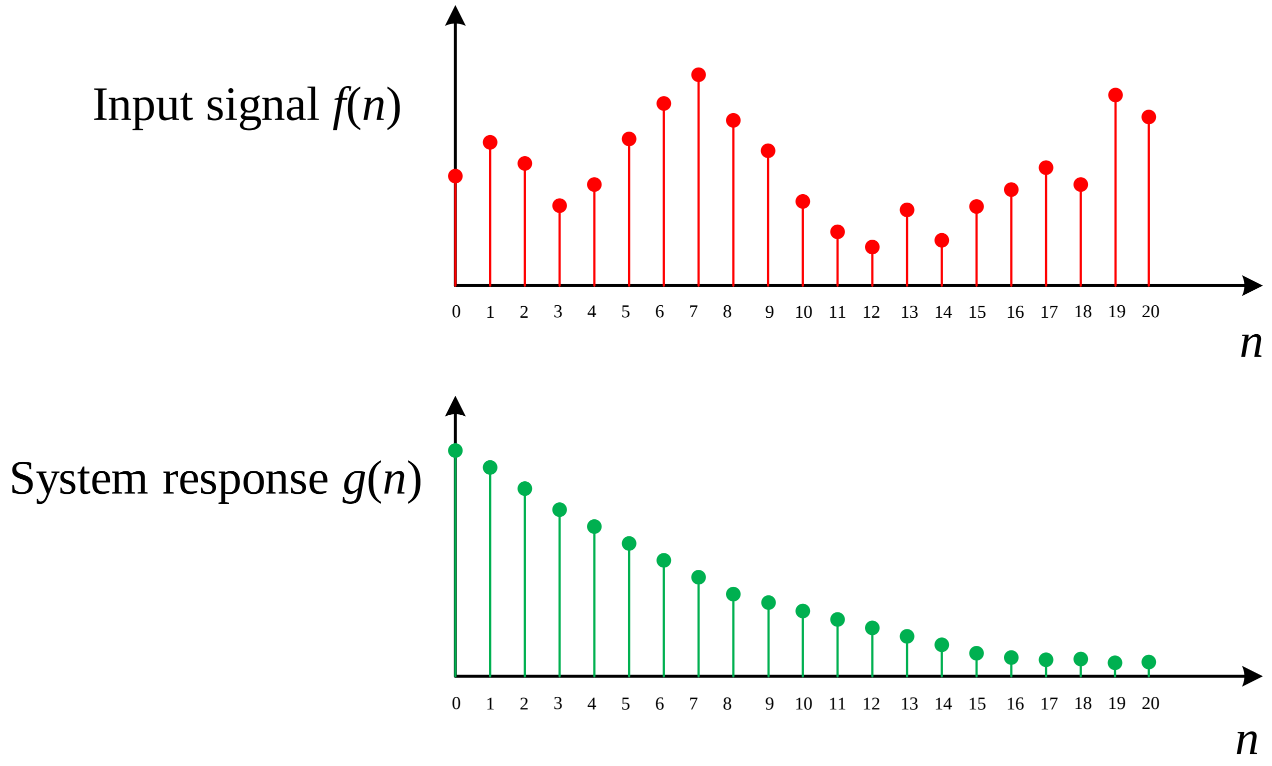
When $|a| > 1$ the sequence diverges; when $|a| < 1$ the sequence converges.



6.7.1 Definition and Physical Meaning of Convolution Sum

The convolution sum of any signals $x_1(n)$ and $x_2(n)$ is defined as:

$$x(n) = x_1(n) * x_2(n) = \sum_{m=-\infty}^{\infty} x_1(m)x_2(n-m)$$

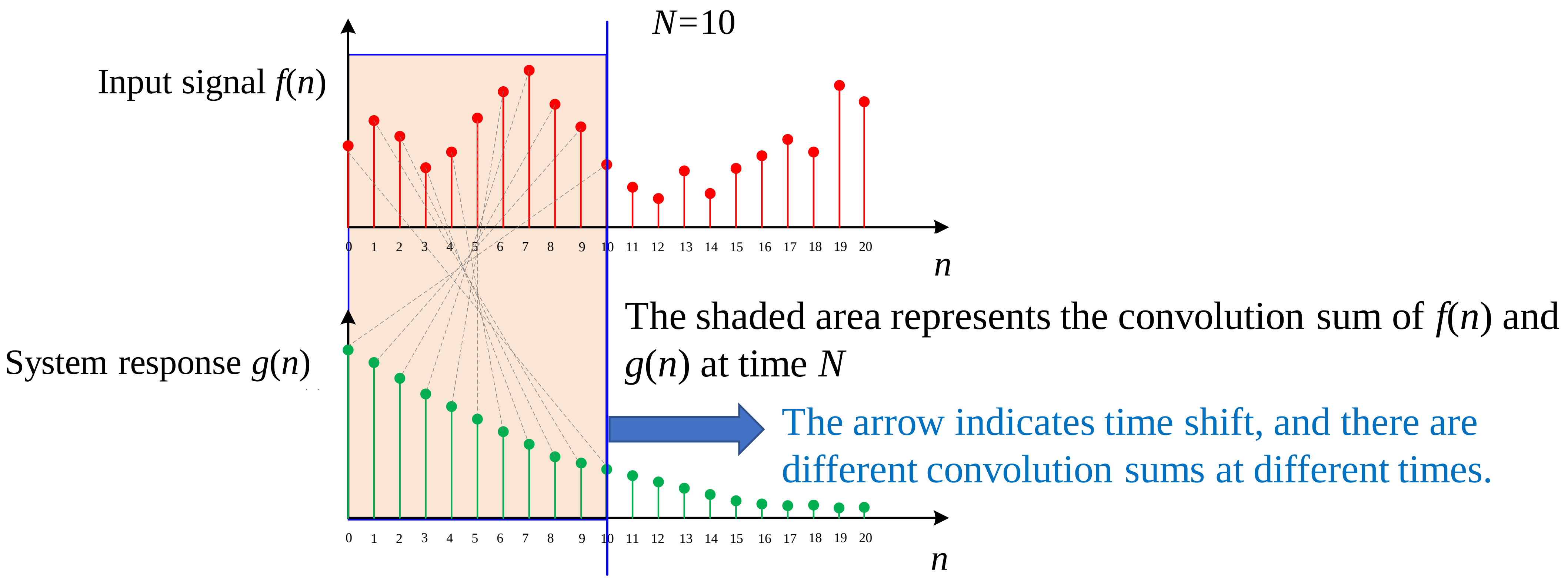


$f(n)$—Input signal.

$g(n)$—Represents the attenuation trend of the input signal value at a certain moment.

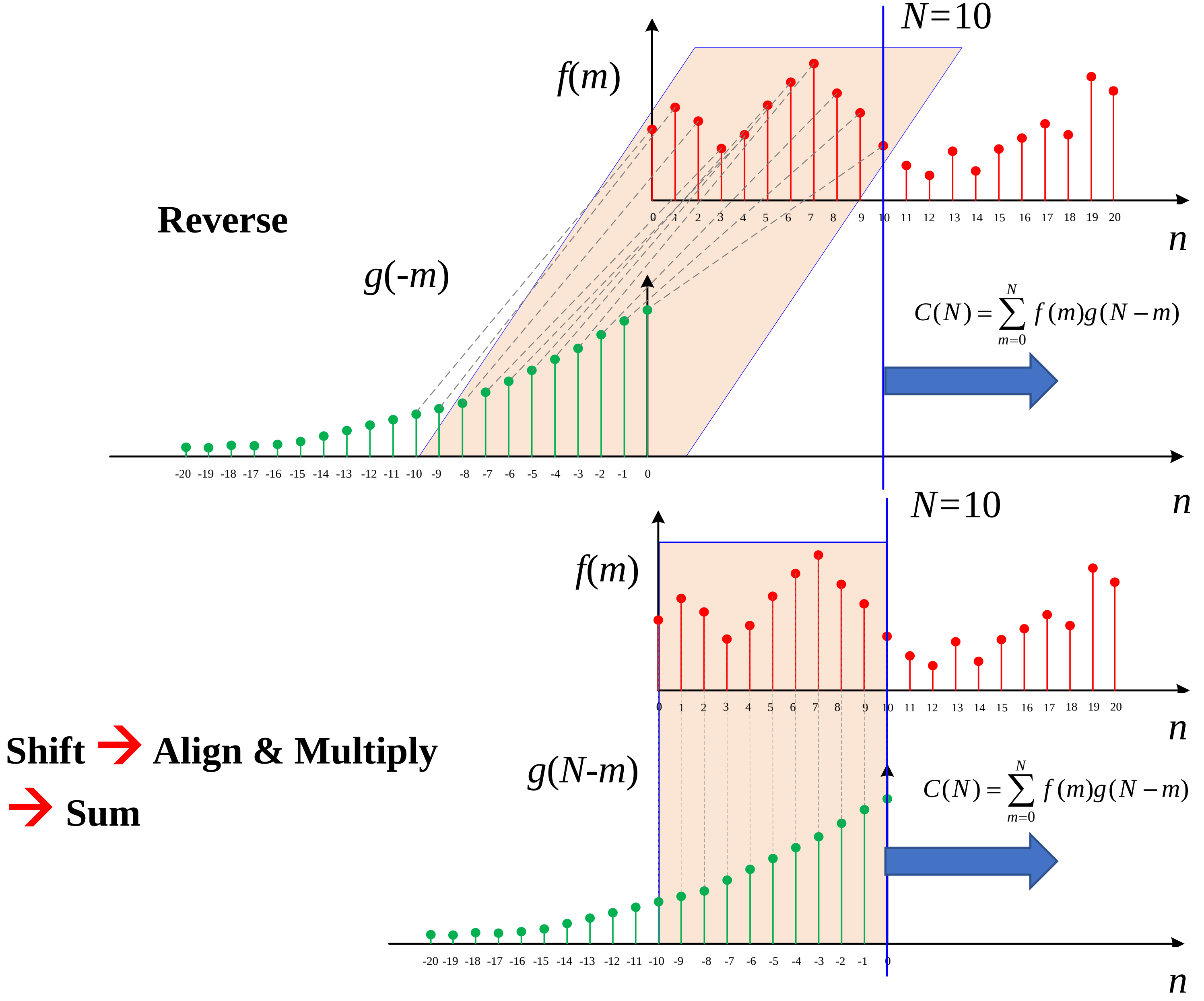
At time N, the input $f(0)$ at n=0 attenuates to $f(0)g(N)$.

6.7 Convolution Sum



The final output of the system reflects the **cumulative effect** of all previous input signals. At ***N*=10**， the output is related to the entire **shaded area**. Since ***f*(10)** is just input, its contribution is ***f*(10)*g*(0)**； ***f*(9)** has attenuated for 1 time unit, contributing ***f*(9)*g*(1)** and so on. Summing these products ***f*(10)*g*(0)+ *f*(9)*g*(1)+… *f*(0)*g*(10)** gives the output at ***N*=10,** which is the **convolution sum** of ***f*(*n*)** and ***g*(*n*)** at N=10**.**

6.7 Convolution Sum



Core Steps of Convolution (The "Four-Step Process")

1. **Reverse**: Transform $g(m)$ to $g(-m)$;
2. **Shift**: Shift $g(-m)$ by t to get $g(n-m)$;
3. **Align and Multiply**: Multiply the overlapping segments of $f(m)$ and $g(N-m)$;
4. **Sum**: Compute the sum of the product over m to get the convolution value at time N.

6.7.2 Finding Zero-State Response Using Convolution Sum

Decompose the excitation signal into a linear combination of unit sample sequences. Using the properties of LTI systems, find the response of each sample sequence acting on the system individually, and superimpose these responses to obtain the **zero-state response** under any excitation.

Any excitation can be expressed as a weighted sum of unit samples:

$$x(n)=\sum_{m=-\infty}^{\infty}x(m)\delta(n-m)$$

$\delta(n)\Rightarrow h(n)$ From time-invariance: $\delta(n-m)\Rightarrow h(n-m)$

From linearity: $y(n)=T\left\{\sum_{m=-\infty}^{\infty}x(m)\delta(n-m)\right\}=\sum_{m=-\infty}^{\infty}x(m)h(n-m)$

$$=\sum_{m=-\infty}^{\infty}h(m)x(n-m)=x(n)*h(n)$$

The zero-state response of a discrete-time LTI system is equal to the convolution sum of the excitation signal and the unit sample response.

6.7.3 Operation Method of Convolution Sum

$$y(n) = x(n) * h(n) = \sum_{m=-\infty}^{\infty} x(m)h(n-m)$$

The graphical method of convolution sum consists of five steps:

(1) Variable Substitution: Change the independent variable in ***x*(*n*)** and ***h*(*n*)** from ***n*** to ***m*,** where ***m*** becomes the independent variable of the function.

(2) Reversing: Fold one of the signals (usually the simpler one)

$$h(m) \xrightarrow{\text{Reverse}} h(-m)$$

(3) Shifting: Shift the folded signal

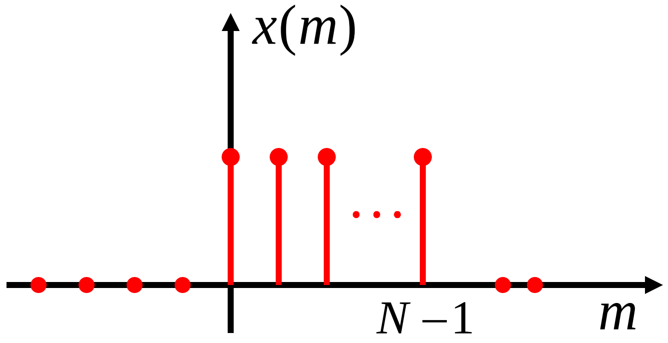
$$h(-m) \xrightarrow{\text{shift by } n} h(-(m-n)) = h(n-m)$$

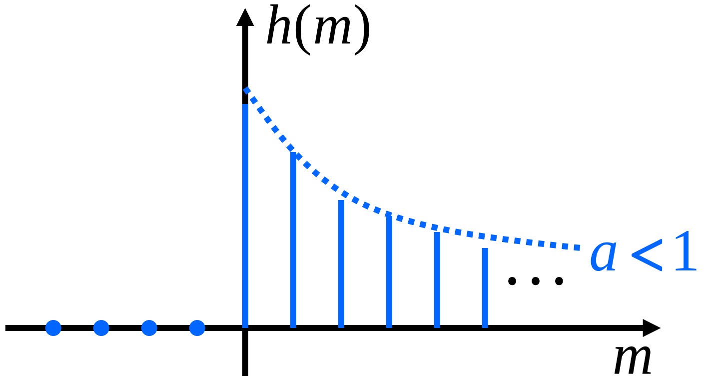
(4) Multiplying: Multiply ***x*(*m*)** by ***h*(*n-m*)**

(5) Summing: Sum the product graph.

Example 6-8: Given the unit sample response of a system $h(n) = a^n u(n)$ where $0 < a < 1$, and the excitation signal $x(n) = R_N(n) = u(n) - u(n-N)$, find the zero-state response.

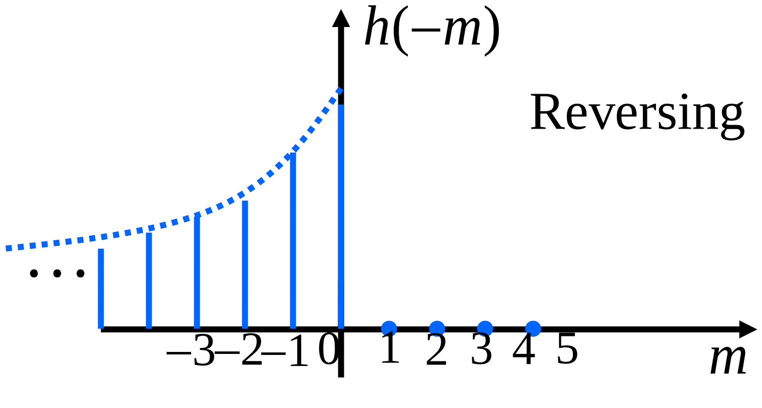
Solution:

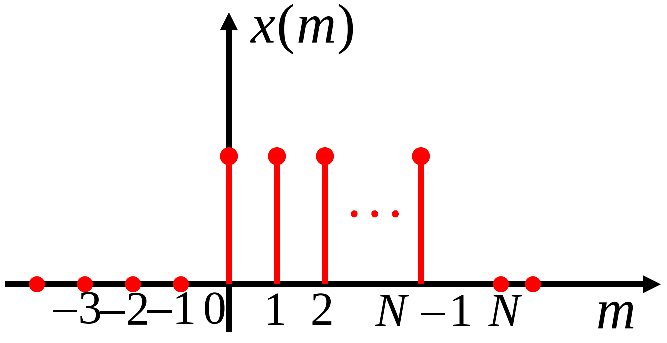




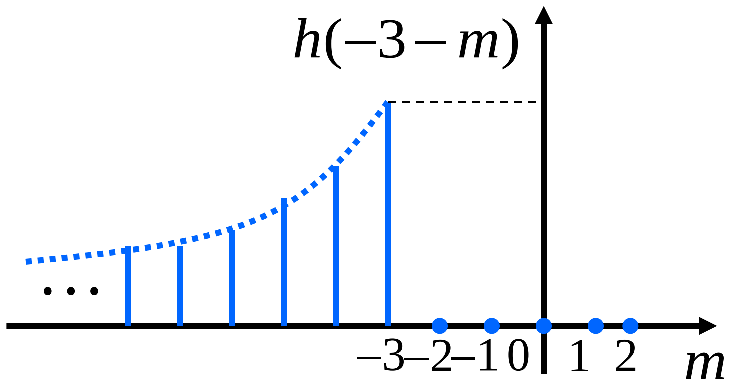
$$y_{zs}(n) = \sum_{m=-\infty}^{\infty} x(m)h(n-m) = \sum_{m=-\infty}^{\infty} [u(m) - u(m-N)]a^{n-m}u(n-m)$$

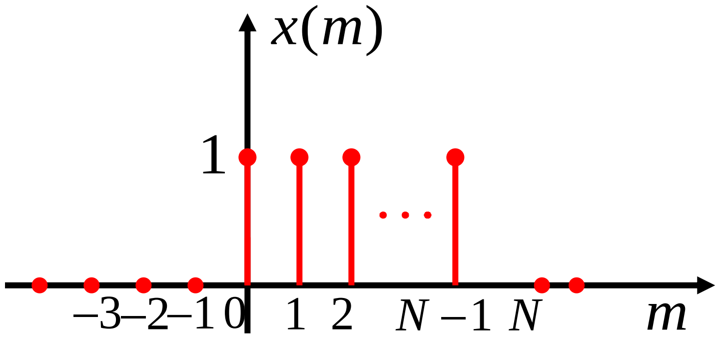
6.7 Convolution Sum





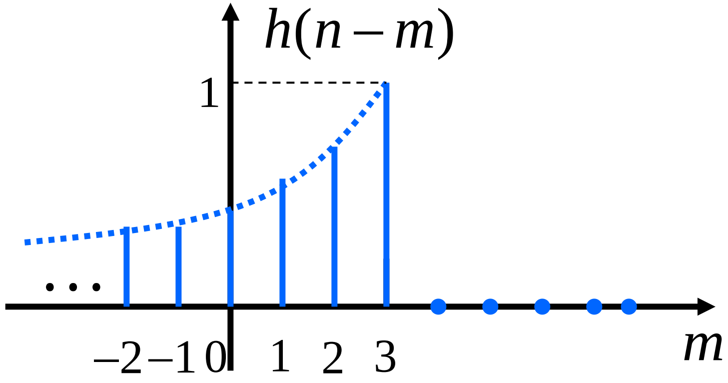
(1)when $n<0$, $x(m)$ and $h(n-m)$ do not overlap, so $y_{zs}(n)=0\ (n<0)$

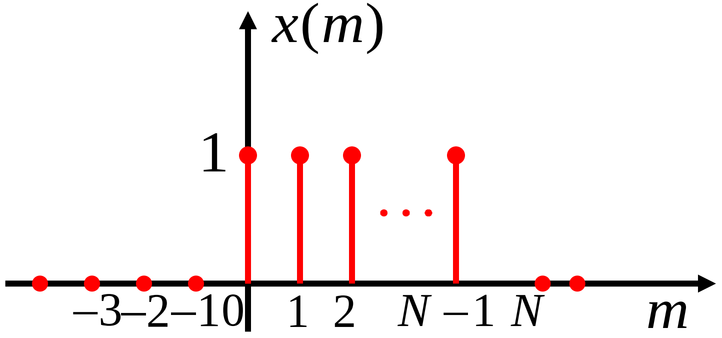




(2)When $0 \le n < N-1$, $x(m)$ and $h(n-m)$ overlap in the interval 0~n

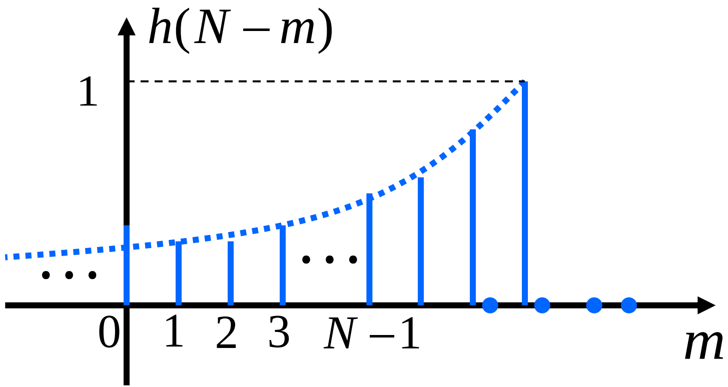
$$y_{zs}(n)=\sum_{m=0}^{n} a^{n-m} = a^{n}\sum_{m=0}^{n} a^{-m} = a^{n}\frac{1-a^{-(n+1)}}{1-a^{-1}} \quad (0 \le n < N-1)$$

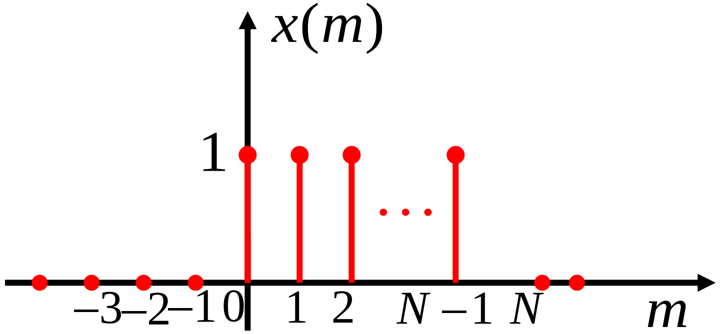




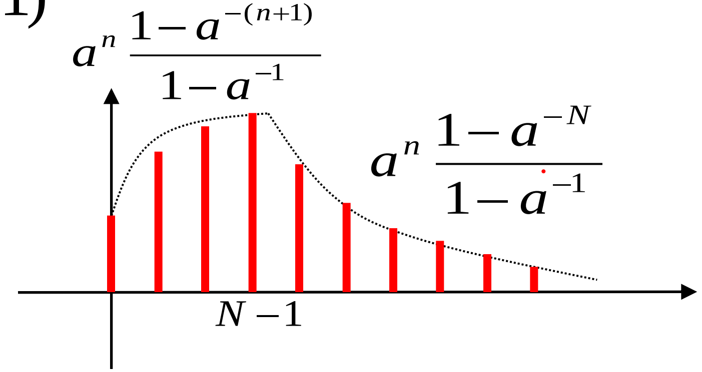
(3) When $n \ge N-1$, $x(m)$ and $h(n-m)$ overlap in the interval 0~$(N-1)$

$$y_{zs}(n)=\sum_{m=0}^{N\text{-}1} a^{n-m} = a^{n}\frac{1-a^{-N}}{1-a^{-1}} \quad \left(n \ge N-1\right)$$





The result of convolution sum



Theorem: Given a discrete-time LTI system, its unit sample response ***h*(*n*)** has non-zero values only in the interval $N_0 \le n \le N_1$, and the input ***x*(*n*)** has non-zero values only in $N_2 \le n \le N_3$. Then, the zero-state response has non-zero values only in $N_4 \le n \le N_5$, where

$$\boxed{N_4 = N_0 + N_2 \quad N_5 = N_1 + N_3}$$

Proof: Let $y(n) = h(n) * x(n) = \sum_{m=-\infty}^{\infty} h(m)x(n-m)$

$$N_0 \le m \le N_1$$

$$N_2 \le n - m \le N_3$$

Add the two inequalities above.

$$N_0 + N_2 \le n \le N_1 + N_3$$

$$N_4 = N_0 + N_2 \quad N_5 = N_1 + N_3$$

6.7 Convolution Sum

Assume the unit sample response ***h*(*n*)** has non-zero values only in $N_0 \le n \le N_1$, with sequence length $L_1 = N_1 - N_0 + 1$. The input ***x*(*n*)** has non-zero values only in $N_2 \le n \le N_3$, with sequence length $L_2 = N_3 - N_2 + 1$.

The zero-state response ***y*(*n*)** has non-zero values only in $N_0 + N_2 \le n \le N_1 + N_3$, with **sequence length *L*$_3$**:

$$L_3 = (N_1 + N_3) - (N_0 + N_2) + 1$$
$$= (N_1 - N_0) + (N_3 - N_2) + 1$$
$$= (N_1 - N_0 + 1) + (N_3 - N_2 + 1) - 1$$

$$\boxed{L_3 = L_1 + L_2 - 1}$$

For continuous-time signals, the length of the convolution integral is $L_3 = L_1 + L_2$.

Example 6-9: Given $x_1(n) = 2\delta(n) + \delta(n-1) + 4\delta(n-2) + \delta(n-3)$

$$x_2(n) = 3\delta(n) + \delta(n-1) + 5\delta(n-2)$$

Find the convolution $y(n) = x_1(n) * x_2(n)$.

Solution: $\{x_1(n)\} = \{\underset{\uparrow}{2} \quad 1 \quad 4 \quad 1\}$ $\qquad \{x_2(n)\} = \{\underset{\uparrow}{3} \quad 1 \quad 5\}$

$$\begin{array}{lrrrrrr}
x_1(n): & & & 2 & 1 & 4 & 1 \\
x_2(n): & & & & 3 & 1 & 5 \\
\hline
& & & 10 & 5 & 20 & 5 \\
& & 2 & 1 & 4 & 1 & \\
& 6 & 3 & 12 & 3 & & \\
\hline
y(n): & 6 & 5 & 23 & 12 & 21 & 5
\end{array}$$

Perform the term-by-term multiplication and summation

no carryover during multiplication and summation

$$\{y(n)\} = \{\underset{\uparrow}{6} \quad 5 \quad 23 \quad 12 \quad 21 \quad 5\}$$

单选题 1分

设置

Calculate the convolution sum of $x(n)=\{1,2,\overset{\downarrow}{0},3,2\}$ and $h(n)=\{1,\overset{\downarrow}{4},2,3\}$

A $y(n)=\{1,6,\overset{\downarrow}{10},10,20,14,13,6\}$

B $y(n)=\{1,8,\overset{\downarrow}{10},10,21,14,13,6\}$

C $y(n)=\{1,6,10,\overset{\downarrow}{10},20,14,13,6\}$

D $y(n)=\{1,8,10,\overset{\downarrow}{10},21,14,13,6\}$

提交

6.7.4 Properties of Convolution Sum

Commutative Law

$$x_1(n)*x_2(n)=x_2(n)*x_1(n)$$

Distributive Law

$$x_1(n)*\left[x_2(n)+x_3(n)\right]=x_1(n)*x_2(n)+x_1(n)*x_3(n)$$

Associative Law

$$x_1(n)*\left[x_2(n)*x_3(n)\right]=\left[x_1(n)*x_2(n)\right]*x_3(n)$$

Convolution with Unit Sample Sequence

$$x(n)*\delta(n)=x(n)$$

$$x(n)*\delta(n-n_0)=x(n-n_0)$$

Convolution with Step Sequence

$$x(n)*u(n)=\sum_{m=-\infty}^{n}x(m)$$

6.7.5 Finding Zero-State Response of a System Using Convolution Sum

Example 6-10: The difference equation of a discrete-time LTI system is

$$y(n)+3y(n-1)+2y(n-2)=x(n)$$

Given the excitation $x(n)=3\left(\frac{1}{2}\right)^n u(n)$**, find the zero-state response $\boldsymbol{y_{zs}(n)}$.**

Solution:

Step 1: Find the unit sample response of the system

$$h(n)+3h(n-1)+2h(n-2)=\delta(n)$$

The characteristic roots are **-1 & -2**, $\therefore h(n)=\left[C_1(-1)^n+C_2(-2)^n\right]u(n)$

$h(-1)=0,\ h(-2)=0,$ iterate to find $h(0)=1.$

Using $h(0)$ and $h(-1)$, get $C_1=-1,\ \ C_2=2.$

$$\therefore h(n)=\left[-(-1)^n+2(-2)^n\right]u(n)$$

Step 2: Find the convolution sum of the excitation signal and the unit sample response to obtain the zero-state response.

$$y_{zs}(n)=\sum_{m=-\infty}^{\infty}x(m)h(n-m)$$

$$=\sum_{m=-\infty}^{\infty}3(\frac{1}{2})^{m}u(m)\cdot[-(-1)^{n-m}+2(-2)^{n-m}]u(n-m)$$

$$=\begin{cases}-3(-1)^{n}\sum_{m=0}^{n}(-\frac{1}{2})^{m}+6(-2)^{n}\sum_{m=0}^{n}(-\frac{1}{4})^{m}, & n\geq 0\\ 0 & n<0\end{cases}$$

$$=[-2(-1)^{n}+\frac{24}{5}(-2)^{n}+\frac{1}{5}(\frac{1}{2})^{n}]u(n)$$

Homework

1. Repeat all the examples in the lecture notes.
2. 2.32(a), 2.34(b), 2.49(f)(g)(i)(j) (judge only causality and stability in 2.49) in the textbook.